

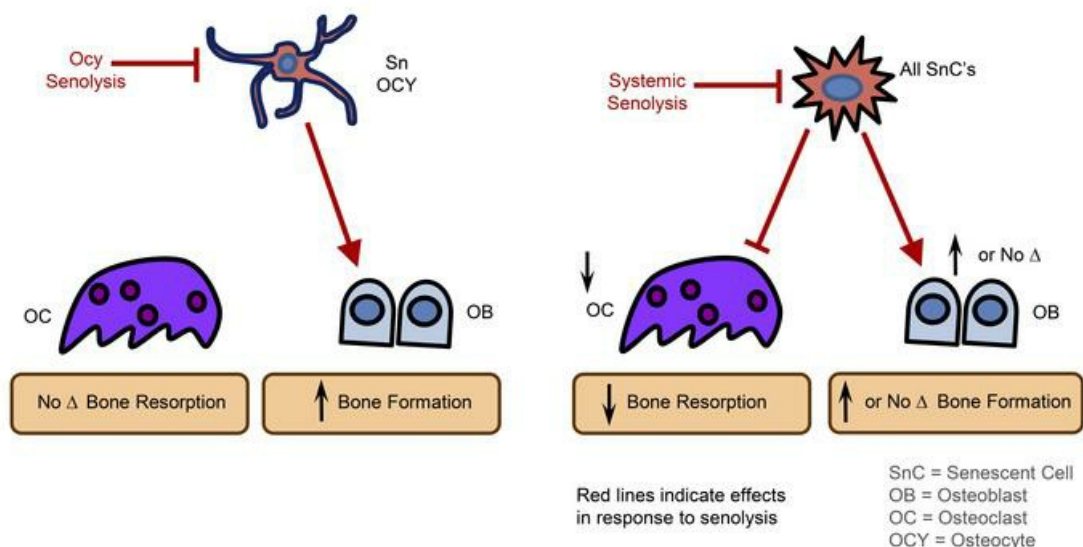
Local senolysis in aged mice only partially replicates the benefits of systemic senolysis

Joshua N. Farr, ... , David G. Monroe, Sundeep Khosla

J Clin Invest. 2023. <https://doi.org/10.1172/JCI162519>.

Research In-Press Preview Aging Bone Biology

Graphical abstract



Find the latest version:

<https://jci.me/162519/pdf>



Local senolysis in aged mice only partially replicates the benefits of systemic senolysis

Joshua N. Farr,^{1,2,3*} Dominik Saul,^{1,2} Madison L. Doolittle,^{1,2} Japneet Kaur,^{1,2} Jennifer L. Rowsey,^{1,2} Stephanie J. Vos,^{1,2} Mitchell N. Froemming,^{1,2} Anthony B. Lagnado,^{1,3} Yi Zhu,^{1,3} Megan Weivoda,⁴ Yuji Ikeno,⁵ Robert J. Pignolo,^{1,3,6} Laura J. Niedernhofer,⁷ Paul D. Robbins,⁷ Diana Jurk,^{1,3} João F. Passos,^{1,3} Nathan K. LeBrasseur,^{1,3,8} Tamara Tchkonja,³ James L. Kirkland,³ David G. Monroe,^{1,2} and Sundeep Khosla^{1,2,3*}

¹Robert and Arlene Kogod Center on Aging, Mayo Clinic College of Medicine, Rochester, Minnesota, USA. ²Division of Endocrinology, Mayo Clinic College of Medicine, Rochester, Minnesota, USA. ³Department of Physiology and Biomedical Engineering, Mayo Clinic College of Medicine, Rochester, Minnesota, USA. ⁴Department of Hematology, Mayo Clinic College of Medicine, Rochester, Minnesota, USA. ⁵Department of Pathology and Laboratory Medicine, University of Texas Health Science Center, San Antonio, Texas, USA. ⁶Department of Medicine, Mayo Clinic College of Medicine, Rochester, Minnesota, USA. ⁷Institute on the Biology of Aging and Metabolism, Department of Biochemistry, Molecular Biology and Biophysics, University of Minnesota, Minneapolis, Minnesota, USA. ⁸Department of Physical Medicine and Rehabilitation, Mayo Clinic College of Medicine, Rochester, MN, USA.

Running title: Local versus systemic senolysis

Keywords: Bone disease; Osteoporosis; Cellular senescence

Corresponding Authors:

*Joshua N. Farr, Ph.D., Guggenheim 7, Mayo Clinic College of Medicine, 200 First Street SW, Rochester, MN 55905; Tel. 507-538-0085; Email: farr.joshua@mayo.edu

*Sundeep Khosla, M.D., Guggenheim 7, Mayo Clinic College of Medicine, 200 First Street SW, Rochester, MN 55905; Tel: +507-255-6663; Email: khosla.sundeep@mayo.edu

Conflict of Interest: Patents on *p16-INK-ATTAC* mice and their uses are held by Mayo Clinic. This research has been reviewed by the Mayo Clinic Conflict of Interest Review Board and was conducted in compliance with Mayo Clinic Conflict of Interest policies.

Abstract

Clearance of senescent cells (SnCs) can prevent several age-related pathologies, including bone loss. However, the local versus systemic roles of SnCs in mediating tissue dysfunction remain unclear. Thus, we developed a mouse model (*p16-LOX-ATTAC*) that allows for inducible SnC elimination (senolysis) in a cell-specific manner and compared the effects of local versus systemic senolysis during aging using bone as a prototype tissue. Specific removal of Sn osteocytes prevented age-related bone loss at the spine, but not the femur, by improving bone formation without affecting osteoclasts or marrow adipocytes. By contrast, systemic senolysis prevented bone loss at the spine and femur and not only improved bone formation, but also reduced osteoclasts and marrow adipocytes. Transplantation of SnCs into the peritoneal cavity of young mice caused bone loss and also induced senescence in distant host osteocytes. Collectively, our findings provide the first proof-of-concept evidence that local senolysis has health benefits in the context of aging, but importantly, local senolysis only partially replicates the benefits of systemic senolysis. Further, we establish that SnCs, through their SASP, lead to senescence in distant cells. Therefore, our study indicates that optimizing senolytic drugs may require systemic instead of local SnC targeting to extend healthy aging.

Introduction

The aging population and life expectancy continue to increase (1), and while unlikely to reverse course, this trend of more people living longer creates challenges society remains poorly equipped to handle, including a myriad of late-life chronic diseases and geriatric syndromes, which cluster in older individuals (2). However, rather than extending lifespan at all costs, compressing the period of late-life morbidity would not only have tremendous economic value (3), but would allow elderly people to maintain autonomy, independence and well-being in old age (4). Because hallmarks of aging promote age-related diseases (5, 6), therapeutically interfering with fundamental aging mechanisms represents an attractive strategy to compress late-life morbidity (7).

In this context, cellular senescence, a permanent state of cell cycle arrest induced by various stressors to halt the proliferation of damaged or dysfunctional cells, has emerged as a key target for therapeutic exploitation to prevent age-related pathologies (8). Senescent cells (SnCs) develop an inflammatory and tissue destructive secretome to prompt their immune-mediated clearance, but which can also drive chronic sterile inflammation, loss of tissue homeostasis and cause aging and age-related diseases (8). Transgenic mouse models were used to demonstrate that elimination of SnCs (*i.e.* senolysis) is therapeutic, able to extend healthspan during chronological aging and reduces the severity of multiple age-related pathologies (9, 10). These findings inspired the discovery of a class of drugs known as senolytics that selectively kill SnCs as a therapeutic approach to extend healthy aging (11). However, as emerging data continue to reinforce the premise that a single senolytic may not be capable of systemically clearing all SnC types (12), it has become of paramount importance to understand the local versus systemic effects of eliminating SnCs. Indeed, elucidating the cell autonomous versus non-autonomous roles of specific SnCs in mediating tissue dysfunction will help guide the development of senolytics (or their combinations) that most broadly alleviate age-related diseases as a group.

In the present study, we hypothesized that bone would arguably represent the ideal tissue in which to address this issue, with a focus on the osteocyte, for multiple reasons: (i) unlike the heterogeneity of most tissues, bone is abundantly populated by osteocytes (comprising >95% of all skeletal cells) (13); (ii) spatial and temporal control of precise gene manipulations in osteocytes using the *Cre-loxP* system is facilitated by a well-characterized *Cre*-line (*i.e.* 8kb *DMP1-Cre*) (14); and (iii) along with a host of other crucial functions (15), the osteocyte operates in a network to precisely control the actions of other skeletal cells, including osteoclasts and osteoblasts, to execute bone remodeling.

Indeed, signals arising from osteocytes, acting *via* the RankL/OPG and Sost/Dkk1/Wnt axes, normally maintain the balance between bone resorption and formation (15). However, these tightly coordinated events go awry in old age through mechanisms that are not fully understood, leading to a net imbalance between bone resorption and bone formation (16). Multipotent mesenchymal stem cells (MSCs) are the source of osteoblasts; however, due to cell autonomous aging mechanisms or in response to non-autonomous aging signals derived from unknown sources, *e.g.* dysfunctional osteocytes (17), these progenitors preferentially differentiate into bone marrow adipocytes in old age (16). Therefore, defective bone formation and increased marrow adiposity are hallmarks of age-related osteoporosis.

Aging bone contains cells with markers of senescence which, as noted earlier, is a stress response program activated by cyclin-dependent kinase inhibitors, most notably *p16^{Ink4a}* (encoded by *Cdkn2a*) and *p21^{Cip1}* (encoded by *Cdkn1a*) (18, 19). With aging, SnCs have increased expression of *p16^{Ink4a}* in several mammalian tissues (20), whereas *p21^{Cip1}* seems to be induced by cells under acute duress leading to senescence in other circumstances (21), including tissue repair (22, 23). Additional molecular properties of SnCs include DNA double-strand breaks at sites of telomeres (*i.e.* telomere-associated foci [TAF] (24)) and, in response to this permanent damage, an upregulated transcriptional machinery to generate a robust secretome comprised of numerous biologically active molecules known as the senescence-

associated secretory phenotype (SASP). These and other senescence markers have been utilized by multiple groups to identify SnCs in the bone microenvironment with aging (25, 26). To establish whether systemic senolysis prevents age-related bone loss, our group eliminated $p16^{Ink4a}+$ SnCs using *p16-INK-ATTAC* (*Apoptosis Through Targeted Activation of Caspase 8*) – a ‘suicide’ transgene driven the $p16^{Ink4a}$ promoter (9). Activation of the $p16^{Ink4a}$ -driven suicide transgene over a 4-month period in old mice prevented age-related trabecular and cortical bone loss at the spine and femur by reducing bone resorption, improving bone formation, and suppressing bone marrow adiposity (27). Thus, systemic clearance of $p16^{Ink4a}+$ SnCs has both anti-resorptive and anabolic effects on aging bone, along with reducing bone marrow adipocytes (28). However, the precise roles of bone-resident versus non-skeletal SnCs in mediating age-related bone loss remain unclear.

To permit the investigation of cell-specific SnC elimination, here we developed and utilized a mouse model (*p16-LOX-ATTAC*) harboring a suicide transgene that allows for inducible senolysis in a cell-specific manner. By comparing local (cell-specific) to systemic senolysis in transgenic mouse models, and through complementary SnC transplantation studies, we dissected the cell autonomous versus non-autonomous effects of SnCs using the skeleton as a prototype tissue.

Results

Sn osteocytes accumulate in bone with advancing age. As an extension of our previous studies (26, 27), we first verified that Sn osteocytes accumulate throughout the lifespan of C57BL/6 WT mice up to age 24 months, a time when age-related bone loss in both female and male mice is well established. Mice were sacrificed at 1, 6, 12, 18, and 24 months of age and highly enriched cell preparations of osteocytes isolated (see Methods) at each timepoint. RT-qPCR analysis confirmed that $p16^{Ink4a}$ transcript levels increased in murine osteocyte-enriched bone samples with advancing age in both sexes when analyzing females and males separately, albeit at somewhat different rates (Figure 1A) as well as in both sexes combined (Supplemental Figure 1A). In addition, as previously shown by our group (26), osteocyte-enriched bones from old mice displayed significant upregulation of SASP components, including chemokines, inflammatory cytokines, and matrix metalloproteases (Supplemental Figure 1B). To establish senescence at the single cell level, we performed TAF staining (Figure 1B), a measure of DNA damage (γ -H2AX) co-localized with telomeres, in osteocytes because TAF are perhaps the most definitive marker of cellular senescence (24). This assay revealed a highly significant accumulation of senescent osteocytes with aging in old (24-month) as compared to young adult (6-month) mice (Figure 1C-E). Given that osteocytes are crucial sources of signaling factors in the coordination of bone remodeling (15), and based on our findings here that at least a subset of osteocytes undergoes senescence and develops a robust SASP with aging, we hypothesized that Sn osteocytes are key drivers of age-related bone loss. To test this hypothesis, we developed and utilized a mouse model, $p16$ -LOX-ATTAC, which allows for the inducible elimination of specific SnC types when crossed with a cell-specific Cre-recombinase (Cre) model – e.g. *8kb DMP1-Cre* for osteocytes (14).

Development and validation of a model for eliminating specific SnC types. In order to permit tissue-specific expression of ATTAC (FKBP-Casp8) (29), the same suicide transgene driven by

the $p16^{Ink4a}$ promoter in $p16$ -*INK*-*ATTAC* mice (9, 10), activated upon administration of a synthetic drug, AP20187 (herein referred to as AP), was placed downstream of a $p16^{Ink4a}$ promoter-*EGFP* cassette with 3 SV40 poly(A) transcription termination sequences (3x*STOP*) flanked by *loxP* sites (Figure 1F). In the un-recombined state, mice harboring this transgene express *EGFP*, but not the FLAG-tagged FKBP-Casp8 fusion protein. However, when crossed with a cell-specific *Cre*, the *EGFP* and 3x*STOP* cassette are removed, resulting in expression of the FLAG-tagged FKBP-Casp8 fusion protein. Thus, in cells with active *Cre*, there is no *EGFP* expression, but the FLAG-tagged FKBP-Casp8 transgene is expressed. Administration of AP to $p16$ -*LOX*-*ATTAC* mice carrying a cell-type specific *Cre* driver leads to activation of the FKBP-Casp8 apoptosis cascade in a specific SnC type to allow for their inducible 'suicide'. Importantly, the combination of cell-specific *Cre* expression and AP permits both cell-specific and temporal (e.g. in aged mice) control of SnC elimination.

To first validate this model, we isolated bone marrow stromal cells (BMSCs) from $p16$ -*LOX*-*ATTAC* mice (un-recombined; Supplemental Figure 2A) or $p16$ -*LOX*-*ATTAC* mice crossed with the ubiquitous *CMV-Cre* (Supplemental Figure 2B), which deletes the *loxP*-flanked gene segments (*EGFP*;3x*STOP*) in all cell types (30). BMSCs isolated from either the un-recombined ($p16$ -*LOX*-*ATTAC*) or recombined (*CMV-Cre*^{+/+}; $p16$ -*LOX*-*ATTAC*) mice were then placed in culture and irradiated (10 Gy) to induce in vitro senescence, which resulted in robust *EGFP* mRNA expression driven by the $p16^{Ink4a}$ promoter in un-recombined Sn BMSCs (Supplemental Figure 2C). In contrast, *EGFP* transcript levels were not detected (all cycle threshold [Ct] values = 40) in Sn BMSCs from *CMV-Cre*^{+/+}; $p16$ -*LOX*-*ATTAC* mice (Supplemental Figure 2C), thus confirming *EGFP* excision. Consistent with this, IHC staining of FLAG, which tags the FKBP-Casp8 portion of the construct, revealed that the FKBP-Casp8 protein was not expressed in Sn BMSCs from $p16$ -*LOX*-*ATTAC* mice unless the cells were derived from $p16$ -*LOX*-*ATTAC* mice crossed with a *Cre* (e.g. *CMV-Cre*; Supplemental Figure 2D).

To validate this model in vivo, we crossed *p16-LOX-ATTAC* mice with *E2a-Cre* mice, which targets expression of *Cre* to the early mouse embryo thus causing *Cre*-mediated recombination in a wide range of tissues (31). We next employed a well-established drug-inducible model (*i.e.* doxorubicin [DoxR]) of cellular senescence in vivo (32, 33). We randomized 4-month-old young adult *E2a-Cre^{+/+};p16-LOX-ATTAC* mice to one of three groups: (i) Vehicle (Veh); (ii) DoxR (10 mg/kg) + Veh; or (iii) DoxR (10 mg/kg) + AP (10 mg/kg, twice weekly) for 24 days (Supplemental Figure 2E). As an indicator of senescence induction, DoxR treatment increased *p16^{Ink4a}* mRNA expression in liver tissue, whereas AP was sufficient to counteract this effect (Supplemental Figure 2F), thus establishing in vivo functionality.

Specific elimination of Sn osteocytes in old mice. To specifically eliminate Sn osteocytes in old age, we crossed *p16-LOX-ATTAC* mice with *8kb DMP1-Cre* mice (14) and aged the female and male *DMP1-Cre^{+/+};p16-LOX-ATTAC* mice to 20 months (baseline). The mice were then randomized to either Veh or AP (10 mg/kg) intraperitoneal (i.p.) injections (twice weekly) for 4-months until 24 months of age (Figure 1G). Compared to Veh, 4-months of AP treatment significantly reduced *p16^{Ink4a}* mRNA expression in osteocyte-enriched bones of both female and male *DMP1-Cre^{+/+};p16-LOX-ATTAC* mice but had no effects on *p16^{Ink4a}* transcript levels in other tissues (Figure 1H), indicating efficient osteocyte-specific senolysis. Results were similar in both females and males, when analyzed separately (Supplemental Figure 3A, B). Osteocytes of *DMP1-Cre^{+/+};p16-LOX-ATTAC* mice treated with 4-months of AP had significantly lower TAF levels compared to Veh-treated mice (Figure 1I), providing confirmation of Sn osteocyte clearance. Further analyses revealed no significant difference in osteoblast TAF levels in AP- versus Veh-treated mice (Figure 1J). However, given that the *8kbDmp-1-Cre* can be active in late osteoblasts (34), and the similar (albeit non-significant in osteoblasts) pattern of TAF reductions in osteocytes and osteoblasts, we cannot fully exclude some effect of our interventions in clearing not only senescent osteocytes but also potentially senescent late

osteoblasts. Finally, staining of senescence-associated β -galactosidase (SA- β -Gal)+ adipocytes in *DMP1-Cre^{+/-};p16-LOX-ATTAC* mice revealed no SnC clearance in adipose tissue (Figure 1K, L), thus providing further evidence that only Sn osteocytes were eliminated in response to AP treatment.

Effects of osteocyte senolysis on body composition and the skeleton in old mice. To assess the impact of clearing Sn osteocytes on body composition and age-related bone loss, we performed echo-magnetic resonance imaging (echo-MRI), dual-energy X-ray absorptiometry (DXA), micro-computed tomography (μ CT), and measurements of bone histomorphometry. As noted above, AP or Veh was delivered i.p. to *DMP1-Cre^{+/-};p16-LOX-ATTAC* female and male mice from 20 to 24 months of age (Figure 2A, I). Mice receiving AP exhibited no changes in body weight at baseline and monthly thereafter when compared to Veh-treated mice (Supplemental Figure 4A-D). Similarly, total body fat mass (Supplemental Figure 4E, F) and lean mass (Supplemental Figure 4G, H) did not differ between AP- and Veh-treated mice at baseline (20-months) or study endpoint (24-months). Furthermore, no effects of AP versus Veh were observed on any body composition parameters when females (Supplemental Figure 5A-H) and males (Supplemental Figure 6A-H) were analyzed separately.

Note that for the primary study endpoints (μ CT outcomes in the different mouse models), the data are shown separately in females and males (including both the Main and Supplemental Figures). In addition, for each of the μ CT outcomes when combining males and females, we performed ANOVA statistical models testing for a sex $\times$ treatment interaction, as recommended by Garcia-Sifuentes and Maney (35) (all $P > 0.05$, Supplemental Table 1). Female mice were randomized to either Veh or AP (10 mg/kg) i.p. injections (twice weekly) for 4-months until 24 months of age (Figure 2A). AP- and Veh-treated female mice were well-matched at baseline (20 months) for lumbar spine DXA-derived areal bone mineral density (aBMD; Figure 2B).

Compared to Veh, Sn osteocyte-specific clearance in response to 4-months of AP treatment in old female mice significantly improved lumbar spine trabecular bone volume fraction (BV/TV; Figure 2C). By contrast, clearance of Sn osteocytes in female mice had no effects on femur trabecular BV/TV (Figure 2D), femur cortical bone parameters (Figure 2E-G), or bone strength (failure load, assessed by micro-finite element analysis [μ FEA]; Figure 2H). Consistent with results in females, AP- and Veh-treated male mice (Figure 2I) were well-matched at baseline (20 months) for lumbar spine DXA-derived aBMD (Figure 2J). Further, compared to Veh, Sn osteocyte-specific clearance in response to 4-months of AP treatment in old male mice significantly improved lumbar spine trabecular BV/TV (Figure 2K), whereas clearance of Sn osteocytes in male mice had no effects on femur trabecular BV/TV (Figure 2L), femur cortical bone parameters (Figure 2M-O), or bone strength (failure load assessed by μ FEA (Figure 2P).

When combining female and male mice (Figure 3A), following ANOVA testing for sex $\times$ treatment interactions (all $P > 0.05$, Supplemental Table 1), AP- and Veh-treated mice were again well-matched at baseline (20 months) for lumbar spine DXA-derived aBMD (Figure 3B). Compared to Veh, Sn osteocyte-specific clearance in response to 4-months of AP treatment significantly improved lumbar spine trabecular bone volume fraction (BV/TV; Figure 3C, D) and bone strength (failure load, assessed by micro-finite element analysis [μ FEA]; Figure 3E) by significantly increasing bone formation rates (BFRs) on spine trabecular surfaces (Figure 3F, G). However, clearance of Sn osteocytes in AP-treated mice had no effect on spine osteoclast (Figure 3H) or osteoblast (Figure 3I) numbers. The increase in spine BFR without an increase in osteoblast numbers would indicate an increase in osteoblast activity accounting for the increase in BFR. We also measured the serum bone formation marker, amino-terminal propeptide of type 1 collagen (P1NP), which did not differ between groups (Veh, 1.39 ± 0.36 vs. AP, 1.62 ± 0.19 , $P=0.352$), likely due to insufficient sensitivity of this marker to detect changes in trabecular BFR in the absence of changes in endocortical BFR along the much more prevalent cortical bone (see below). At the femur metaphysis, Sn osteocyte clearance in AP-treated mice did not affect

femur BV/TV (Figure 3J) and only resulted in a modest, albeit non-significant ($P=0.078$), increase in cortical thickness (Figure 3K) without affecting endocortical (Figure 3L) or periosteal (Figure 3M) circumferences. Collectively, 4-months of AP treatment in old *DMP1-Cre^{+/-};p16-LOX-ATTAC* mice was not sufficient to significantly change femoral bone strength (μ FEA-derived failure load; Figure 3N). However, Sn osteocyte clearance in AP-treated mice did significantly improve BFRs on periosteal (Figure 3O, P), but not endocortical (Figure 3Q), surfaces of the femur metaphysis, thus explaining the modest improvement in cortical thickness observed in AP-treated mice. Clearance of Sn osteocytes in AP-treated mice had no effect on femoral osteoclast (Figure 3R), osteoblast (Figure 3S), or marrow adipocyte (Figure 3T) numbers.

In addition to aging the *DMP1-Cre^{+/-};p16-LOX-ATTAC* mouse colony, as a control we also aged a separate cohort of *p16-LOX-ATTAC* mice not crossed with a *Cre* (non-crossed controls) to 20 months of age and delivered AP or Veh i.p. (10 mg/kg, twice weekly) to females and males for 4-months (Supplemental Figures 7-9). Aged female *p16-LOX-ATTAC* (non-crossed) mice did not differ at baseline (20 month) for spine aBMD (Supplemental Figure 7B). Further, after 4-months of Veh- or AP-treatment, there were no differences in spine trabecular parameters (Supplemental Figure 7C-F), femur trabecular BV/TV (Supplemental Figure 7G), femur cortical parameters (Supplemental Figure 7H-K), or femoral bone strength (μ FEA-derived failure load; Supplemental Figure 7L) between old female *p16-LOX-ATTAC* (non-crossed) Veh- and AP-treated mice. Results were similar in aged male *p16-LOX-ATTAC* (non-crossed) Veh- and AP-treated mice where no between-group differences in bone parameters were observed (Supplemental Figure 8A-L). In addition when combining sexes, following ANOVA testing for sex $\times$ treatment interactions (Supplemental Table 1), both groups were well-matched at baseline (20 months) for spine aBMD (Supplemental Figure 9B) and 4-months of AP treatment had no significant effects on μ CT-derived bone parameters at the lumbar spine (Supplemental Figure 9C-F) or femur metaphysis (Supplemental Figure 9G-L), thus demonstrating that the

ATTAC portion of the *p16-LOX-ATTAC* transgene is not activated by AP if these mice are not crossed with a *Cre* model (*i.e.* there is no 'leakiness' of the *p16-LOX-ATTAC* transgene).

Finally, we aged another separate cohort of *p16-LOX-ATTAC* mice crossed with β -*Actin-Cre* mice, which results in widespread tissue targeting of *Cre* (36), and aged the resulting β -*Actin-Cre*^{+/+};*p16-LOX-ATTAC* mice to 20 months when AP or Veh was delivered i.p. (10 mg/kg, twice weekly) to females and males for 4-months (Supplemental Figure 10A). Consistent with previously published findings in response to systemic SnC clearance in old *p16-INK-ATTAC* mice (10, 27, 37), AP treatment in old β -*Actin-Cre*^{+/+};*p16-LOX-ATTAC* mice resulted in significantly reduced *p16*^{ink4a} mRNA expression in multiple tissues, including osteocyte-enriched bones, fat, kidney, and liver (Supplemental Figure 10B). Similar effects were observed in females and males when analyzed separately, although due to the variability of the *p16*^{ink4a} mRNA measurements, some differences between the sexes were evident, albeit not significant (all *P*-values >0.05) for a sex $\times$ treatment interaction by ANOVA (Supplemental Figure 10C, D; Supplemental Table 1).

Consistent with Sn osteocyte-specific clearance, systemic SnC clearance in old male β -*Actin-Cre*^{+/+};*p16-LOX-ATTAC* had no significant effects on body weight (Supplemental Figure 11A-D), fat mass (Supplemental Figure 11E-G), or lean mass (Supplemental Figure 11H-J), and while both male groups were well-matched at baseline for spine aBMD (Supplemental Figure 11K), 4-months of AP treatment as compared to Veh in old males resulted in significant improvements in lumbar spine BV/TV (Supplemental Figure 11L), femur BV/TV (Supplemental Figure 11M), femur cortical thickness (Supplemental Figure 11N), and femoral bone strength (μ FEA-derived failure load, Supplemental Figure 11O). In old female β -*Actin-Cre*^{+/+};*p16-LOX-ATTAC* mice, results were similar (Supplemental Figure 12A-O); however, improvements in response to AP treatment did not reach statistical significance (*i.e.* using parametric statistics) for trabecular BV/TV at the spine (Supplemental Figure 12L; *t*-Test *P*=0.070; Wilcoxon Rank-Sum Test *P*=0.065) or trabecular BV/TV at the femur (Supplemental Figure 12M; *t*-Test

$P=0.138$; Wilcoxon Rank-Sum Test $P=0.034$) due to suboptimal statistical power as a result of higher than anticipated numbers of deaths in the aged β -Actin-Cre^{+/+};p16-LOX-ATTAC cohort. Notwithstanding, despite the relatively smaller sample size (n=11 ♀ Veh vs n=11 ♀ AP) compared to our previously published study in aged female p16-INK-ATTAC mice (n=13 ♀ Veh vs n=16 ♀ AP) where systemic clearance of SnCs significantly improved trabecular bone parameters at both the spine and femur as well as femur cortical bone parameters (27), old female β -Actin-Cre^{+/+};p16-LOX-ATTAC mice treated with AP had significant improvements in femur cortical thickness (Supplemental Figure 12N) and femoral strength (failure load assessed by μ FEA (Supplemental Figure 12O). When combining the aged female and male β -Actin-Cre^{+/+};p16-LOX-ATTAC mice, following ANOVA testing for sex $\times$ treatment interactions (Supplemental Table 1), AP- and Veh-treated mice had similar body composition (Supplemental Figure 13A-J) and were well-matched at baseline for spine aBMD (Supplemental Figure 13K), whereas 4-months of AP treatment as compared to Veh resulted in significant improvements in lumbar spine BV/TV (Supplemental Figure 13L), femur BV/TV (Supplemental Figure 13M), femur cortical thickness (Supplemental Figure 13N), and femoral bone strength (μ FEA-derived failure load, Supplemental Figure 13O). Taken together, these results demonstrate that while some effects of Sn osteocyte-specific clearance on the skeleton were consistent with systemic SnC clearance (e.g. BV/TV at the spine), other effects were quite different than findings in old p16-INK-ATTAC mice, as previously reported by our group (27).

Effects of osteocyte senolysis on the osteocyte network. Because aging causes detrimental changes to the osteocyte lacunocanalicular network (LCN), we next examined the impact of Sn osteocyte-specific clearance on multiple aspects of the osteocyte LCN in old DMP1-Cre^{+/+};p16-LOX-ATTAC mice treated with AP versus Veh for 4-months from 20 to 24 months of age (Figure 4A). Examination of >400 osteocytes per mouse (Figure 4B), revealed that there were no changes in the percentage of osteocyte empty lacunae between the AP- and Veh-treated

groups (Figure 4C, D). Additional analysis of >1000 osteocytes per animal (Figure 4E), suggested a modest increase in the percentage of osteocyte apoptosis assessed by TUNEL assay in the AP- versus Veh-treated mice (Figure 4F, G), although this increase did not reach statistical significance ($p=0.165$, Wilcoxon non-parametric test). Interestingly, while the quality of the osteocyte LCN as reflected by numbers of osteocyte dendrites and their density both naturally degenerate in mice with aging (38), 4-months of AP treatment as compared to Veh in old *DMP1-Cre^{+/-};p16-LOX-ATTAC* mice was sufficient to significantly improve the quality of the osteocyte LCN (Figure 4H; see Methods for a detailed description of the osteocyte LCN score), as assessed by phalloidin staining and confocal imaging (Figure 4I).

Effects of local versus systemic senolysis on the SASP and Sn-related pathways. In order to compare changes in the bone SASP as well as Sn-related genes and to characterize the key underlying intercellular signaling pathways in response to Sn osteocyte-specific (local) versus systemic SnC clearance, we next utilized a gene set (SenMayo), that has been previously validated for identifying SnCs across multiple tissues and species, including mice and humans, with high fidelity (39). SenMayo is a senescence gene set comprised of 117 genes in mice, that consists predominantly of SASP factors ($n=76$) but also includes transmembrane ($n=19$) and intracellular ($n=22$) proteins. In our previous analysis of young versus old mice, we found that SenMayo expression increased with aging, and that systemic SnC clearance (using *p16-INK-ATTAC* mice) following 4-months of AP treatment reduced the expression of multiple SASP and Sn-related genes (39).

Here we used SenMayo to examine earlier changes in the SASP and Sn-related transcriptome (after only two weeks of AP) in bones of old mice following Sn osteocyte-specific (*DMP1-Cre^{+/-};p16-LOX-ATTAC*) or systemic (*p16-INK-ATTAC*) SnC clearance. Application of SenMayo to whole-bones (*i.e.* not specifically enriched for osteocytes) obtained from 24-month-old *DMP1-Cre^{+/-};p16-LOX-ATTAC* and *p16-INK-ATTAC* mice demonstrated highly significant

downregulation of the SenMayo geneset by AP in both *p16-INK-ATTAC* and *DMP1-Cre^{+/-};p16-LOX-ATTAC* mice (Supplemental Figure 14; MANOVA *P*-values for overall comparison of gene sets [see *Statistics*] <0.005 for both), consistent with a senolytic effect of AP in both models. Next, to directly interrogate genes differentially altered by senolysis in the osteocyte-specific (local) versus systemic SnC clearance models, we compared the SenMayo panel between the AP-treated groups. This demonstrated a large majority of genes that were lower in the systemic as compared to the local model (Figure 5A), consistent with a more profound senolytic effect in response to systemic SnC clearance. Specifically, when considering the individual SenMayo genes that were significantly different between the AP-treated groups, systemic senolysis resulted in 28 genes that were significantly lower relative to local senolysis, whereas local senolysis resulted in only 4 genes that were significantly lower when compared to systemic senolysis (Figure 5B). We next applied BioPlanet2019 pathway analysis (Figure 5C), which revealed the senescence and autophagy as well as the p53/IL2/Egf/Egfr pathways as being most regulated after systemic senolysis and the Wnt/Lrp6/IL10/IL4 pathways as the most regulated following local senolysis.

To further extend these findings, we next measured an array of cytokines and chemokines (total of 32 protein targets) in serum from these old mice. With local senolysis, only a single target (IP10) was reduced, whereas ten targets (GMCSF, IFNG, IL13, IL17, MCP1, MIP1A, MIP2B, RANTES, TNFa, and VEGF) were reduced in response to systemic senolysis, although when considering the entire array using multivariate analysis of variance (MANOVA), between-group differences were not significant following Sn osteocyte-specific (*DMP1-Cre^{+/-};p16-LOX-ATTAC*; MANOVA *P*-value = 0.247) or systemic (*p16-INK-ATTAC*; MANOVA *P*-value = 0.509) SnC clearance (Supplemental Figure 15A, B). Importantly, however, when the array of cytokines and chemokines was restricted to only those targets included in our recently validated senescence gene set (SenMayo, (39)), whereas changes in these targets remained non-significant (MANOVA *P*-value = 0.744) following local senolysis (Supplemental Figure 16A),

systemic senolysis resulted in a significant (MANOVA P -value = 0.045) reduction in SenMayo proteins, including IL13, MCP1, MIP1A, MIP1B, RANTES, TNFa, and VEGF (Supplemental Figure 16B). Collectively, these data indicate that systemic senolysis has greater effects on circulating inflammatory cytokines as compared to osteocyte-specific senolysis and moreover, the reduction in inflammatory factors following systemic senolysis is relatively specific for known SASP factors.

Finally, in addition to overlap as well as differences in SASP/Sn-related genes and pathways, further RT-qPCR analyses revealed that *Sost* (encoding for sclerostin) levels in whole-bones were significantly reduced with both systemic and local senolysis (Figure 5D), whereas only with systemic senolysis, but not with local senolysis, was *Rankl* (also known as *Tnfsf11*) significantly reduced in whole-bones of old AP- relative to Veh-treated mice (Figure 5E). These findings are entirely consistent with the increase in bone formation observed in both models, but a decrease in bone resorption occurring only in the systemic senolysis model.

Transplanting SnCs into young mice leads to skeletal aging. To test whether non-skeletal SnCs can induce aging bone phenotypes, we first developed a reproducible protocol for isolating primary fibroblasts from young adult C57BL/6 mice and making them Sn in culture. Indeed, in vitro exposure of primary murine fibroblasts to 10Gy irradiation (IR), followed by a 20-day waiting period, caused most surviving IR-exposed cells (relative to non-IR, Control cells) to become Sn using criteria based on the combination of SA- β -Gal⁺ staining and markedly increased mRNA expression of both *p16^{Ink4a}* and *p21^{Cip1}* (Figure 6A, B). We previously found that the SASP of IR-induced Sn murine fibroblasts is consistent with that of in vivo SnCs with aging (40, 41), and using luciferase tracking that these transplanted SnCs remain in the peritoneal cavity following transplantation by i.p. injection and do not travel to distant organs including bone (42).

We next transplanted Sn or non-Sn (Control) murine primary fibroblasts ($\sim 10^6$) isolated from young adult C57BL/6 WT mice into young adult (4-month-old), syngeneic male WT mice, and then waited 2-months prior to sacrificing these animals and subsequently performed extensive skeletal phenotyping (Figure 6C). At 6-months of age, μ CT of the lumbar spine revealed that mice transplanted with $\sim 10^6$ SnCs versus similar numbers of non-Sn Control cells had significantly lower trabecular BV/TV (Figure 6D). Similarly, SnC transplantation caused cortical thinning and deficient bone strength (*i.e.* μ FEA-derived failure load) at the femur metaphysis (Figure 6E, F). SnC transplantation significantly reduced circulating levels of the bone formation marker, serum amino-terminal propeptide of type I collagen (P1NP), in young mice transplanted with SnCs versus non-Sn Control cells (Figure 6G), but had no effect on circulating levels of the bone resorption marker, serum C-terminal telopeptide of type I collagen (CTX; Figure 6H). Consistent with these findings in the circulation, bone histomorphometry revealed that both osteoblast numbers (Figure 6I) and BFRs (Figure 6J, K) were significantly reduced in young mice transplanted with SnCs, but there were no changes in osteoclast numbers in SnC transplanted as compared with non-Sn Control cell transplanted young mice (Figure 6L). In contrast with the deficient numbers of osteoblasts, there was a marked increase in marrow adipocytes in SnC transplanted mice as compared to young mice transplanted with non-Sn Control cells (Figure 6M, N). Finally, because our group has previously shown that transplanted cells (both Sn and non-Sn Control) remain detectable *in vivo* for only up to 40 days post transplantation (42), we next asked at 60 days after SnC transplantation, beyond the time these cells are no longer detectable, whether the transplanted SnCs induced Sn in normal, previously healthy host cells *in vivo*. Indeed, two months after transplantation, TAF staining revealed significantly more TAF⁺ osteocytes (Figure 6O), but no difference in osteoblast TAF levels (Figure 6P), in SnC transplanted mice as compared to mice transplanted with non-Sn Control cells, thus demonstrating that senescence had spread to distant host osteocytes.

The SASP impairs bone formation and alters MSC lineage commitment. Because SnC clearance in aged mice improves bone formation and inhibits marrow adiposity and in youth SnC transplantation causes premature age-related effects on osteoblast and marrow adipocyte lineage commitment, we hypothesized that the SASP produced by SnCs is causal in the pathogenesis of the age-related defect in bone formation and alteration of MSC lineage commitment. To test this hypothesis experimentally, we induced in vitro senescence of murine primary fibroblasts (as above), and 20 days post-IR, collected the SnC conditioned medium (CM) (Figure 7A), which is a rich source of SASP factors (40-42). We then cultured BMSCs isolated from young adult C57BL/6 WT mice under osteogenic conditions in the presence of either Sn CM or Control CM (Figure 7B). After 14 days, Sn CM markedly impaired the mineralization of BMSCs as compared to Control CM (Figure 7C, D). Consistent with this, Sn CM treatment reduced alkaline phosphatase (AP) staining in BMSCs treated with Sn CM versus Control CM (Figure 7E). In addition, RT-qPCR analyses revealed significantly reduced mRNA levels of classic bone marker genes (*i.e. Alpl, Bglap, Runx2*) during osteoblastic differentiation of BMSCs exposed to Sn CM versus Control CM (Figure 7F). Taken together, these data support a key mechanistic link between the SASP produced by SnCs and impaired bone formation, which occurs in vivo both with aging and in response to SnC transplantation in younger mice, and which can be rescued in old mice by both Sn osteocyte-specific and systemic SnC clearance. Further, because systemic, but not osteocyte-specific, clearance of SnCs results in reduced marrow adiposity, which can be induced in young mice transplanted with SnCs, we lastly complemented these in vivo findings with in vitro studies, showing that treatment of primary murine BMSCs isolated from young C57BL/6 WT mice and cultured under adipogenic conditions in the presence of fibroblast Sn CM resulted in increased adipocyte formation as compared to Control CM (Figure 7G), and was associated with increased mRNA expression of multiple adipogenic (*i.e. Adipoq, Fabp4, Pparg*) markers (Figure 7H).

Discussion

Because multiple lines of evidence reinforce the premise that a single senolytic may not be capable of systemically clearing all SnC types (12), it has become of paramount importance to understand the local versus systemic effects of eliminating SnCs. Indeed, elucidating the cell autonomous versus non-autonomous roles of specific SnCs in mediating tissue dysfunction will help guide the development of senolytics (or their combinations) that most broadly alleviate age-related diseases as a group. Here, we developed and utilized a transgenic mouse that allows for the inducible removal of specific types of SnCs. Our results establish that local senolysis in old age only partially replicates the benefits of systemic senolysis. These results were complemented by SnC transplantation studies, demonstrating that the SASP from SnCs in the peritoneal cavity can induce senescence systemically, in distant host osteocytes. Collectively, in addition to dissecting cell autonomous versus non-autonomous mechanisms of senescence, our studies indicate that optimizing senolytic drugs may require systemic “broad spectrum” SnC targeting to extend healthy aging.

We had previously demonstrated that systemic SnC clearance using the *p16-INK-ATTAC* model prevented age-related bone loss in mice (27). Here we used a mouse model (*p16-LOX-ATTAC*) to specifically clear Sn osteocytes, which resulted in substantial, but incomplete, skeletal benefits when compared to systemic SnC clearance using either the *p16-INK-ATTAC* model or the *p16-LOX-ATTAC* mice crossed with a ubiquitously-expressed *Cre* (β -*Actin*). We believe that the “cleanest” controls for the *DMP1-Cre^{+/+};p16-LOX-ATTAC* mice are the *p16-INK-ATTAC* mice as in this model systemic SnC clearance is assured because this construct is simply driven by the *p16^{Ink4a}* promoter and does not depend on uniform expression of a “tissue non-specific” *Cre*. Specifically, although the β -*actin-Cre* is considered a ubiquitous *Cre*, it likely is not equally active across all tissues due both to the specific promoter and perhaps the integration site of the transgene. We acknowledge, however, that our choice of the *p16-INK-ATTAC* mice as the control in some experiments does raise the possibility that the

minimally different transgene constructs between the *DMP1-Cre^{+/+};p16-LOX-ATTAC* and the *p16-LOX-ATTAC* could be a potential confounder.

While we found that the specific removal of Sn osteocytes in old mice was sufficient to prevent trabecular bone loss at the spine by improving bone formation, it did not alter femur trabecular bone parameters, only modestly enhanced cortical thickness by improving periosteal (but not endocortical) bone formation and had no effects on bone resorption or marrow adiposity. The increases bone formation at the spine as well as at the periosteal cortex without changes in osteoblast numbers following osteocyte-specific senolysis suggests that synthetic activity (and/or longevity) of periosteal osteoblasts is regulated by senescent osteocytes perhaps via IGFBP5, a component of SenMayo (39), which has been shown previously to suppress bone formation on the periosteum but not endosteum (43, 44). These findings in response to osteocyte-specific SnC clearance contrast with systemic SnC clearance in old *p16-INK-ATTAC* mice, where systemic elimination of *p16^{Ink4a+}* SnCs prevents age-related trabecular and cortical bone loss at the spine and femur by not only increasing bone formation but also by reducing bone resorption, and suppressing bone marrow adiposity (27).

As anticipated, clearance of Sn osteocytes in old mice resulted in significantly reduced TAF⁺ osteocytes; however, it did not alter the percentage of total osteocyte empty lacunae and only modestly increased osteocyte apoptosis. Furthermore, specific elimination of Sn osteocytes significantly improved the quality of the osteocyte LCN as reflected by numbers of osteocyte dendrites and their density. Although these findings may at first seem paradoxical, based on our previous work (26), we know that only a small percentage of osteocytes (~10% or fewer) become Sn with aging, even when studying mice late in life (e.g. at 24-months). Therefore, because AP treatment in *INK-ATTAC* mice does not clear all Sn osteocytes (i.e. only ~30% reduction), this ultimately results in a very limited proportion of osteocytes being eliminated among the overall vast network. Nevertheless, despite only clearing this very small population of dysfunctional osteocytes following 4-months of treatment, there were clear benefits to trabecular

bone at the spine, and our short-term Sn osteocyte clearance study showed profound local SASP reduction (using bone mRNA analyses) following just two weeks, which is likely the key driver of the beneficial skeletal changes observed and may also produce a more permissive environment for osteocytes to refill empty lacunae. Moreover, we hypothesize that suppressing the SASP in the bone *milieu* has additional benefits, beyond the scope of our study, toward improving skeletal health in aging. For example, Sn osteocyte clearance significantly improved the quality of the osteocyte LCN, and given the recent demonstration that the LCN architecture is a primary determinant of the healthy skeletal response to mechanical loads (45), our data suggest that therapeutic strategies to eliminate SnCs and suppress the SASP could represent effective approaches to restore osteocyte function and health in old age.

Although the SASP was suppressed in whole-bones with both local (*DMP*) and systemic (*INK*) senolysis, a direct comparison between the two AP-treated groups (*DMP* versus *INK*) demonstrated a broader downregulation of SASP/Sn-related genes and pathways (e.g. senescence, autophagy, p53 signaling) with systemic SnC clearance. As might be expected, local senolysis in bone predominantly affected the Wnt and several related pathways. In addition, in response SnC clearance in both models, we observed a reduction in *Sost*, indicating that an anti-sclerostin mediated anabolic mechanism leads to increased bone formation following both local (*DMP*) and systemic (*INK*) senolysis. By contrast, *Rankl* was only reduced in whole-bones following systemic SnC clearance, which raises the possibility that other SnC types, in addition to Sn osteocytes, are important sources of RankL in old age, although given the systemic nature of the *p16-INK-ATTAC* model we cannot exclude the possibility that systemic clearance of SnCs through effects on other cell types leads to reduced bone resorption. The latter is supported by experiments showing that the SASP produced by Sn preadipocytes enhances osteoclastogenesis by promoting the survival of osteoclast precursors, which can be blocked pharmacologically using a JAK 1/2 inhibitor and by in vivo studies demonstrating that inhibiting the SASP in old mice prevents bone loss and reduces bone

resorption (27). Nevertheless, our findings establish that Sn osteocytes are clearly not the only culprits mediating skeletal aging.

To definitively establish a systemic role for senescence, we transplanted non-skeletal SnCs (*i.e.* Sn fibroblasts) intraperitoneally into young adult mice, which after two months resulted in hallmarks of skeletal aging, including bone loss, a reduction in osteoblasts, an increase in marrow adiposity, and induction of senescence in osteocytes within the host. While acknowledging that there is complex heterogeneity in the SASP among different SnC types (46), these cell transplantation studies in combination with our *in vitro* mechanistic studies establish ‘proof-of-concept’ that the SASP of non-skeletal SnCs can inhibit osteogenesis while simultaneously enhancing adipogenesis, providing evidence that the SASP has profound effects on MSC lineage allocation in bone. Moreover, while previous *in vitro* studies had shown that SnCs can propagate senescence to neighboring cells in a paracrine manner (47, 48), our study establishes that SnCs can induce senescence *in vivo* in cells in a distant organ (bone) using perhaps the most definitive assay for cellular senescence, *i.e.* TAF (24).

In summary, using both genetic and SnC transplantation models to either eliminate $p16^{Ink4a+}$ Sn cells in old age or induce an aging Sn *milieu* in bone in younger mice, our studies demonstrate that both Sn osteocytes and non-skeletal SnCs contribute to skeletal aging. In contrast with systemic SnC clearance (27), Sn osteocyte clearance in old mice only partially prevents age-related bone loss by improving bone formation through a mechanism that involves both suppression of the SASP and sclerostin. We also find that systemic, but not local osteocyte-specific, SnC clearance reduces RankL, which indicates that other cells, in addition to senescent osteocytes, are important mediators of bone resorption in aging. Further, we demonstrate that the SASP causes an alteration in lineage commitment of MSCs towards the osteoblast and away from the adipocyte lineage, which is reversed with systemic SnC clearance, but not with Sn osteocyte-specific clearance. We complement these genetic approaches with transplantation of senescent cells into the peritoneal cavity of young mice and

demonstrate that not only does this lead to systemic bone loss through the SASP of these cells but remarkably, actually induces senescence of osteocytes within bone, assessed using the most rigorous assay currently available for defining senescent cells, the presence of TAFs. To our knowledge, this is the first evidence of senescent cells inducing “senescence at-a-distance” using these rigorous criteria. Finally, we establish the utility of *p16-LOX-ATTAC* mice, which will be useful in future studies when crossed with other mouse *Cre*-drivers, for dissecting the contributions of additional specific SnC types in mediating age-related tissue dysfunction. Thus, the evidence presented here best fits a model whereby SnCs promote aging through both cell autonomous and non-autonomous effects, therefore establishing that the most effective senolytics (or combinations thereof) will systemically eliminate SnCs to extend healthy aging.

Methods

General experimental approaches. Mice were randomly assigned to experimental groups, as specified below. Equal numbers of mice representing each sex were balanced between treatment groups. No mice or samples were excluded from analyses, which were conducted in a blinded fashion.

Mice. As detailed here and in the Figure Legends, various mouse models (both females and males), all in the *C57BL/6* background, were used in experiments including the following: *C57BL/6 Wild-Type (WT)*, *p16-INK-ATTAC* (9, 10), and *p16-LOX-ATTAC* (see below for details). All mice, including *Cre*-recombinase (*Cre*) mice, were maintained in the *C57BL/6* background. The *p16-LOX-ATTAC* mice were either non-crossed (controls) or crossed with *DMP1-Cre* (for mature osteoblasts and osteocytes (14) mice as well as *CMV-Cre* (ubiquitous expression (30)), *E2a-Cre* (ubiquitous expression (31)), or β -*Actin-Cre* (ubiquitous expression (36)), which target expression of *Cre* to the early mouse embryo thus causing *Cre*-mediated recombination in various tissues. Mice were housed in ventilated cages within an accredited facility under a 12 hr light/dark cycle with constant temperature (23°C) and had access to H₂O and food (standard mouse chow, Lab Diet 5053, St. Louis, MO), *ad libitum*.

p16-LOX-ATTAC mouse model. In order to achieve cell-specific expression of *ATTAC* and thereby enable the conditional elimination of specific SnC types, Dr. David G. Monroe (Mayo Clinic) developed a construct where the same *p16^{Ink4a}* promoter (devised by Wang *et al.* (49)) used in the *p16-INK-ATTAC* model (9, 10) drives expression of a lox-stop-lox (LSL) cassette that includes EGFP and three SV40 poly(A) transcription termination sequences (3xSTOP) flanked by *loxP* sites, followed by the *ATTAC* cassette (devised by P. Scherer (29)) that encodes a FLAG-tagged FKBP-Casp8 fusion protein. This construct was made by first digesting the attB-containing pBT378 plasmid (50) with HindIII and NotI and cloning in a linker containing additional SbfI and NsiI sites for subsequent cloning steps. The mouse *p16^{Ink4a}* promoter, encompassing -2617 to +81 relative to the transcription start site, was amplified using LongAmp

Taq DNA polymerase (New England Biolabs, Ipswich, MA) and cloned into the HindIII site. Next, the LSL cassette was produced as a gBLOCK (Integrated DNA Technologies [IDT], Coralville, IA) and cloned into the 3' HindIII and SbfI sites. Finally, the *ATTAC* cassette, followed by a bGH-polyA signal, was produced as a gBLOCK and cloned into the SbfI and NsiI sites. Transgenic mice were produced through the Stanford Transgenic, Knockout and Tumor Model Center by selectively inserting the *p16-LOX-ATTAC* construct into the site-specific Rosa26 locus in C57BL/6 mice using integrase-mediated transgenesis (50). This technique ensures high efficiency single-copy transgene insertion into a predetermined and transcriptionally active chromosomal locus. In the un-recombined state (no Cre), transgene harboring mice express *EGFP* but not the FLAG-tagged FKBP-Caspase 8 (Casp8) fusion protein; however, when crossed with a *Cre* model, the *EGFP* and 3xSTOP cassette are removed, resulting in expression of the FLAG-tagged FKBP-Casp8 fusion protein. Thus, in cells with an active *Cre*-recombinase, there is no *EGFP* expression, but the FLAG-tagged FKBP-Casp8 transgene is expressed. Administration of AP20187 to *LOX-ATTAC* mice crossed with a *Cre* model prompts activation of the FKBP-Casp8 apoptosis cascade in a specific SnC type (*Cre*-dependent) to allow for their inducible 'suicide'. Importantly, the combination of the *Cre* and AP20187 permits us to attain both cell- specific and temporal (in old mice after onset of age-related bone loss) control of SnC elimination.

Doxorubicin (DoxR)-induced senescence in vivo study. This study is described in detail in the Supplemental Methods.

Mouse harvests and tissue collections. Mice were harvested and tissues were collected as previously described by our group (27), and as detailed in the Supplemental Methods.

Biochemical assays. All biochemical assays were performed in blinded fashion, as detailed in Supplemental Methods.

Multiplex protein analyses. Multiplexing analysis was performed using the Luminex 100 system (Luminex) by Eve Technologies Corp., as detailed in Supplemental Methods.

Body composition assessments. All body composition assessments were performed on mice that were non-anesthetized and conscious, as detailed in Supplemental Methods.

Skeletal assessments. Skeletal imaging. All imaging and analysis was performed in a blinded fashion as described by our group previously (27), and as detailed in Supplemental Methods.

Skeletal histomorphometry assessments. All bone histomorphometry analyses were performed in blinded fashion as previously described by our group (27), and as detailed in Supplemental Methods.

TUNEL assay to detect apoptotic osteocytes. These methods are described in detail in the Supplemental Methods.

Osteocyte lacunocanicular network (LCN) analysis. These methods are described in detail in the Supplemental Methods.

Conditioned medium generation. As described previously by our group (27), these methods are detailed in Supplemental Methods.

In vitro osteoblast and adipocyte differentiation. These methods are detailed in Supplemental Methods.

Immunostaining for FLAG. These methods are detailed in the Supplemental Methods.

Real-Time quantitative polymerase chain reaction (RT-qPCR) analyses. Details for the RT-qPCR procedures and analyses are provided in Supplemental Methods.

Telomere-Associated Foci (TAF). To measure osteocyte or osteoblast cellular senescence, as described previously (51), the TAF assay was performed on non-decalcified methacrylate-embedded bone sections, as detailed in the Supplemental Methods.

Statistics. Graphical data are shown using dot plots as means $\pm$ SEM unless otherwise specified. No mice or samples were excluded from statistical analyses. Data were examined for distribution and normality (skewness and kurtosis) using histograms and dot plots. When the normality or equal-variance assumptions for parametric analysis methods were not met, data were analyzed using non-parametric statistics (e.g. Wilcoxon's rank-sum test). For parametric tests, as

dependent on the comparison, between-group differences were analyzed using either an independent samples *t*-Test or one-way ANOVA, as statistically justified and specified in Figure Legends. If a statistically significant ($P < 0.05$) effect was determined by ANOVA, pairwise multiple comparisons were corrected as appropriate using the Tukey *post hoc* method. Comparison of pre-specified groups of gene sets or cytokine array data was performed using a multivariate analysis of variance (MANOVA) model as previously validated for such comparisons (52). Group sample sizes were based on previously published experiments in our laboratory (27, 51), where statistically significant ($P < 0.05$) differences were observed for various senescence- and skeletal-related endpoints in response to multiple interventions. For heatmaps, the z-scores were calculated using the dCT values with a transformation according to the formula:

$$Z = \frac{x - \mu}{\sigma} \text{ with } Z = \text{standard score, } x = \text{observed value, } \mu = \text{mean of sample and } \sigma = \text{standard deviation.}$$

To detect significant differences among the 117 SenMayo genes, we chose a student *t*-Test (two-tailed) with a cut-off of $P < 0.05$ (two-tailed) considered statistically significant. The BioVenn diagram was designed as recommended by Hulsen et al. (53), whereas pathway analysis was performed using gene ontology, while choosing the BioPlanet2019 pathways (Enrichr, BioPlanet 2019 (54)) and focusing on pathways with a Benjamini-Hochberg correction for multiple testing of $P_{\text{adj}} < 0.05$. The combined score is calculated by taking the log of each *P*-value and multiplying by the z-score from the expected rank. Group sizes and sex (♀, ♂, or both) are specified for each experiment in corresponding Figure Legends. To eliminate sex as a potential confounder, in analyses where females and males were combined, equal (or as close as possible) numbers of mice representing each sex were balanced between treatment groups. In addition, for the primary endpoints (μ CT outcomes in the different mouse models), the key data are shown separately in females and males (including the Main and Supplemental Figures). As recommended by Garcia-Sifuentes and Maney (35), we have also used the ANOVA model testing for a sex $\times$ treatment interaction, and if this was not significant (P -value > 0.05 , as was the case for all ANOVAs, as

shown in Supplemental Table 1), the pooled data are presented. For the secondary analyses, following a test for a sex $\times$ treatment interaction, the pooled data are presented if the interaction was not significant (P -value >0.05). Statistical analyses were performed using either GraphPad Prism (GraphPad Software Inc., La Jolla, CA, Version 9.3) or the Statistical Package for the Social Sciences for Windows, Version 25.0 (SPSS, Chicago, IL), with a P -value of <0.05 (two-tailed) considered statistically significant.

Study approval. Animal studies were performed under IACUC approved protocols and experiments were executed in accordance with Mayo Clinic IACUC guidelines and regulations.

Supplemental Data. Supplemental Table 1, Supplemental Figures 1-17, and the Supplemental Methods are included as Supplemental Materials.

Author contributions: JNF, DGM, and SK conceived the studies and designed experiments. JNF and SK directed and supervised all aspects of the study and take responsibility for the integrity of the data analysis. JNF, DS, MLD, JK, JLR, SJV, MNF, ABL, and YZ performed experiments. JNF and DS analyzed data. JNF and SK interpreted data. DGM designed the *p16-LOX-ATTAC* transgene construct. JNF wrote the initial manuscript and all authors contributed ideas and participated in editing and revising the manuscript.

Data availability: All data is provided as Source Data or Supplementary Data.

Acknowledgements

Funding: We gratefully acknowledge funding from the NIH: AG062413 (SK, JNF, NKL, JLK, TT), R01 DK128552 (JNF), R01 AG076515 (SK, DGM), R01 AG063707 (DGM), R01 AG068048 (JFP), R37 AG013925 (JLK, TT), 1UG3CA26810 (JFP), R01AG063543 (LJN), U19AG056278 (PDR, LJN), and the Connor Group (JLK, TT), Robert J. and Theresa W. Ryan (JLK, TT), and

the Noaber Foundation (JLK, TT). We would like to thank the Stanford Transgenic, Knockout, and Tumor Model Center for inserting the *p16-LOX-ATTAC* transgene into C57BL/6 mice. JLK., TT, J. M. van Deursen, and D. J. Baker (Mayo Clinic) designed the strategy for and developed *INK-ATTAC* mice.

References

1. Partridge L, Deelen J, and Slagboom PE. Facing up to the global challenges of ageing. *Nature*. 2018;561(7721):45-56.
2. St Sauver JL, Boyd CM, Grossardt BR, Bobo WV, Finney Rutten LJ, Roger VL, et al. Risk of developing multimorbidity across all ages in an historical cohort study: differences by sex and ethnicity. *BMJ Open*. 2015;5(2):e006413.
3. Goldman DP, Cutler D, Rowe JW, Michaud PC, Sullivan J, Peneva D, et al. Substantial health and economic returns from delayed aging may warrant a new focus for medical research. *Health Aff (Millwood)*. 2013;32(10):1698-705.
4. Kirkland JL. Translating advances from the basic biology of aging into clinical application. *Exp Gerontol*. 2013;48(1):1-5.
5. Lopez-Otin C, Blasco MA, Partridge L, Serrano M, and Kroemer G. The hallmarks of aging. *Cell*. 2013;153(6):1194-217.
6. Kennedy BK, Berger SL, Brunet A, Campisi J, Cuervo AM, Epel ES, et al. Geroscience: linking aging to chronic disease. *Cell*. 2014;159(4):709-13.
7. Sierra F, and Kohanski R. Geroscience and the trans-NIH Geroscience Interest Group, GSIG. *Geroscience*. 2017;39(1):1-5.
8. Khosla S, Farr JN, Tchkonja T, and Kirkland JL. The role of cellular senescence in ageing and endocrine disease. *Nat Rev Endocrinol*. 2020;16(5):263-75.
9. Baker DJ, Wijshake T, Tchkonja T, LeBrasseur NK, Childs BG, van de Sluis B, et al. Clearance of p16Ink4a-positive senescent cells delay aging-associated disorders. *Nature*. 2011;479(7372):232-6.
10. Xu M, Palmer AK, Ding H, Weivoda MM, Pirtskhalava T, White TA, et al. Targeting senescent cells enhances adipogenesis and metabolic function in old age. *Elife*. 2015;4:e12997.
11. Zhu Y, Tchkonja T, Pirtskhalava T, Gower AC, Ding H, Giorgadze N, et al. The Achilles' heel of senescent cells: from transcriptome to senolytic drugs. *Aging Cell*. 2015;14(4):644-58.
12. Kirkland JL, and Tchkonja T. Cellular Senescence: A Translational Perspective. *EBioMedicine*. 2017;21:21-8.
13. Bonewald LF. The amazing osteocyte. *J Bone Miner Res*. 2011;26(2):229-38.
14. Bivi N, Condon KW, Allen MR, Farlow N, Passeri G, Brun LR, et al. Cell autonomous requirement of connexin 43 for osteocyte survival: consequences for endocortical resorption and periosteal bone formation. *J Bone Miner Res*. 2012;27(2):374-89.
15. Robling AG, and Bonewald LF. The Osteocyte: New Insights. *Annu Rev Physiol*. 2020;82:485-506.
16. Sfeir JG, Drake MT, Khosla S, and Farr JN. Skeletal Aging. *Mayo Clin Proc*. 2022;97(6):1194-208.
17. Farr JN, Kaur J, Doolittle ML, and Khosla S. Osteocyte Cellular Senescence. *Curr Osteoporos Rep*. 2020;18(5):559-67.
18. Alcorta DA, Xiong Y, Phelps D, Hannon G, Beach D, and Barrett JC. Involvement of the cyclin-dependent kinase inhibitor p16 (INK4a) in replicative senescence of normal human fibroblasts. *Proc Natl Acad Sci U S A*. 1996;93(24):13742-7.
19. Beausejour CM, Krtolica A, Galimi F, Narita M, Lowe SW, Yaswen P, et al. Reversal of human cellular senescence: roles of the p53 and p16 pathways. *Embo J*. 2003;22(16):4212-22.
20. Krishnamurthy J, Torrice C, Ramsey MR, Kovalev GI, Al-Regaiey K, Su L, et al. Ink4a/Arf expression is a biomarker of aging. *J Clin Invest*. 2004;114(9):1299-307.

21. Sturmlechner I, Zhang C, Sine CC, van Deursen EJ, Jeganathan KB, Hamada N, et al. p21 produces a bioactive secretome that places stressed cells under immunosurveillance. *Science*. 2021;374(6567):eabb3420.
22. Saul D, Monroe DG, Rowsey JL, Kosinsky RL, Vos SJ, Doolittle ML, et al. Modulation of fracture healing by the transient accumulation of senescent cells. *Elife*. 2021;10.
23. Chandra A, Lagnado AB, Farr JN, Doolittle M, Tchkonja T, Kirkland JL, et al. Targeted clearance of p21- but not p16-positive senescent cells prevents radiation-induced osteoporosis and increased marrow adiposity. *Aging Cell*. 2022;21(5):e13602.
24. Hewitt G, Jurk D, Marques FD, Correia-Melo C, Hardy T, Gackowska A, et al. Telomeres are favoured targets of a persistent DNA damage response in ageing and stress-induced senescence. *Nat Commun*. 2012;3:708.
25. Piemontese M, Almeida M, Robling AG, Kim HN, Xiong J, Thostenson JD, et al. Old age causes de novo intracortical bone remodeling and porosity in mice. *JCI Insight*. 2017;2(17).
26. Farr JN, Fraser DG, Wang H, Jaehn K, Ogrodnik MB, Weivoda MM, et al. Identification of Senescent Cells in the Bone Microenvironment. *J Bone Miner Res*. 2016;31(11):1920-9.
27. Farr JN, Xu M, Weivoda MM, Monroe DG, Fraser DG, Onken JL, et al. Targeting cellular senescence prevents age-related bone loss in mice. *Nat Med*. 2017;23(9):1072-9.
28. Farr JN, and Khosla S. Cellular senescence in bone. *Bone*. 2019;121:121-33.
29. Pajvani UB, Trujillo ME, Combs TP, Iyengar P, Jelicks L, Roth KA, et al. Fat apoptosis through targeted activation of caspase 8: a new mouse model of inducible and reversible lipoatrophy. *Nat Med*. 2005;11(7):797-803.
30. Schwenk F, Baron U, and Rajewsky K. A cre-transgenic mouse strain for the ubiquitous deletion of loxP-flanked gene segments including deletion in germ cells. *Nucleic Acids Res*. 1995;23(24):5080-1.
31. Lakso M, Pichel JG, Gorman JR, Sauer B, Okamoto Y, Lee E, et al. Efficient in vivo manipulation of mouse genomic sequences at the zygote stage. *Proc Natl Acad Sci U S A*. 1996;93:5860-5.
32. Ewald JA, Desotelle JA, Wilding G, and Jarrard DF. Therapy-induced senescence in cancer. *J Natl Cancer Inst*. 2010;102(20):1536-46.
33. Baar MP, Brandt RMC, Putavet DA, Klein JDD, Derks KWJ, Bourgeois BRM, et al. Targeted Apoptosis of Senescent Cells Restores Tissue Homeostasis in Response to Chemotoxicity and Aging. *Cell*. 2017;169(1):132-47 e16.
34. Kalajic I, Matthews BG, Torreggiani E, Harris MA, Divieti Pajevic P, and Harris SE. In vitro and in vivo approaches to study osteocyte biology. *Bone*. 2013;54(2):296-306.
35. Garcia-Sifuentes Y, and Maney DL. Reporting and misreporting of sex differences in the biological sciences. *Elife*. 2021;10.
36. Lewandoski M, Meyers EN, and Martin GR. Analysis of Fgf8 gene function in vertebrate development. *Cold Spring Harb Symp Quant Biol*. 1997;62:159-68.
37. Lagnado A, Leslie J, Ruchaud-Sparagano MH, Victorelli S, Hirsova P, Ogrodnik M, et al. Neutrophils induce paracrine telomere dysfunction and senescence in ROS-dependent manner. *Embo J*. 2021;40(9):e106048.
38. Tiede-Lewis LM, Xie Y, Hulbert MA, Campos R, Dallas MR, Dusevich V, et al. Degeneration of the osteocyte network in the C57BL/6 mouse model of aging. *Aging (Albany NY)*. 2017;9(10):2190-208.
39. Saul D, Kosinsky RL, Atkinson EJ, Doolittle ML, Zhang X, LeBrasseur NK, et al. A new gene set identifies senescent cells and predicts senescence-associated pathways across tissues. *Nat Commun*. 2022;13(1):4827.

40. Xu M, Tchkonina T, Ding H, Ogrodnik M, Lubbers ER, Pirtskhalava T, et al. JAK inhibition alleviates the cellular senescence-associated secretory phenotype and frailty in old age. *Proc Natl Acad Sci U S A*. 2015;112(46):E6301-10.
41. Xu M, Bradley EW, Weivoda MM, Hwang SM, Pirtskhalava T, Decklever T, et al. Transplanted Senescent Cells Induce an Osteoarthritis-Like Condition in Mice. *J Gerontol A Biol Sci Med Sci*. 2017;72(6):780-5.
42. Xu M, Pirtskhalava T, Farr JN, Weigand BM, Palmer AK, Weivoda MM, et al. Senolytics improve physical function and increase lifespan in old age. *Nat Med*. 2018;24(8):1246-56.
43. Salih DA, Mohan S, Kasukawa Y, Tripathi G, Lovett FA, Anderson NF, et al. Insulin-like growth factor-binding protein-5 induces a gender-related decrease in bone mineral density in transgenic mice. *Endocrinology*. 2005;146(2):931-40.
44. Kim KS, Seu YB, Baek SH, Kim MJ, Kim KJ, Kim JH, et al. Induction of cellular senescence by insulin-like growth factor binding protein-5 through a p53-dependent mechanism. *Mol Biol Cell*. 2007;18(11):4543-52.
45. van Tol AF, Schemenz V, Wagermaier W, Roschger A, Razi H, Vitiennes I, et al. The mechanoreponse of bone is closely related to the osteocyte lacunocanalicular network architecture. *Proc Natl Acad Sci U S A*. 2020;117(51):32251-9.
46. Hernandez-Segura A, de Jong TV, Melov S, Guryev V, Campisi J, and Demaria M. Unmasking Transcriptional Heterogeneity in Senescent Cells. *Curr Biol*. 2017;27(17):2652-60 e4.
47. Nelson G, Wordsworth J, Wang C, Jurk D, Lawless C, Martin-Ruiz C, et al. A senescent cell bystander effect: senescence-induced senescence. *Aging Cell*. 2012;11(2):345-9.
48. Nelson G, Kucheryavenko O, Wordsworth J, and von Zglinicki T. The senescent bystander effect is caused by ROS-activated NF-kappaB signalling. *Mech Ageing Dev*. 2018;170:30-6.
49. Wang W, Wu J, Zhang Z, and Tong T. Characterization of regulatory elements on the promoter region of p16(INK4a) that contribute to overexpression of p16 in senescent fibroblasts. *J Biol Chem*. 2001;276(52):48655-61.
50. Tasic B, Hippenmeyer S, Wang C, Gamboa M, Zong H, Chen-Tsai Y, et al. Site-specific integrase-mediated transgenesis in mice via pronuclear injection. *Proc Natl Acad Sci U S A*. 2011;108(19):7902-7.
51. Eckhardt BA, Rowsey JL, Thicke BS, Fraser DG, O'Grady KL, Bondar OP, et al. Accelerated osteocyte senescence and skeletal fragility in mice with type 2 diabetes. *JCI Insight*. 2020;5(9).
52. Tsai CA, and Chen JJ. Multivariate analysis of variance test for gene set analysis. *Bioinformatics*. 2009;25(7):897-903.
53. Hulsen T, de Vlieg J, and Alkema W. BioVenn - a web application for the comparison and visualization of biological lists using area-proportional Venn diagrams. *BMC Genomics*. 2008;9:488.
54. Chen EY, Tan CM, Kou Y, Duan Q, Wang Z, Meirelles GV, et al. Enrichr: interactive and collaborative HTML5 gene list enrichment analysis tool. *BMC Bioinformatics*. 2013;14:128.

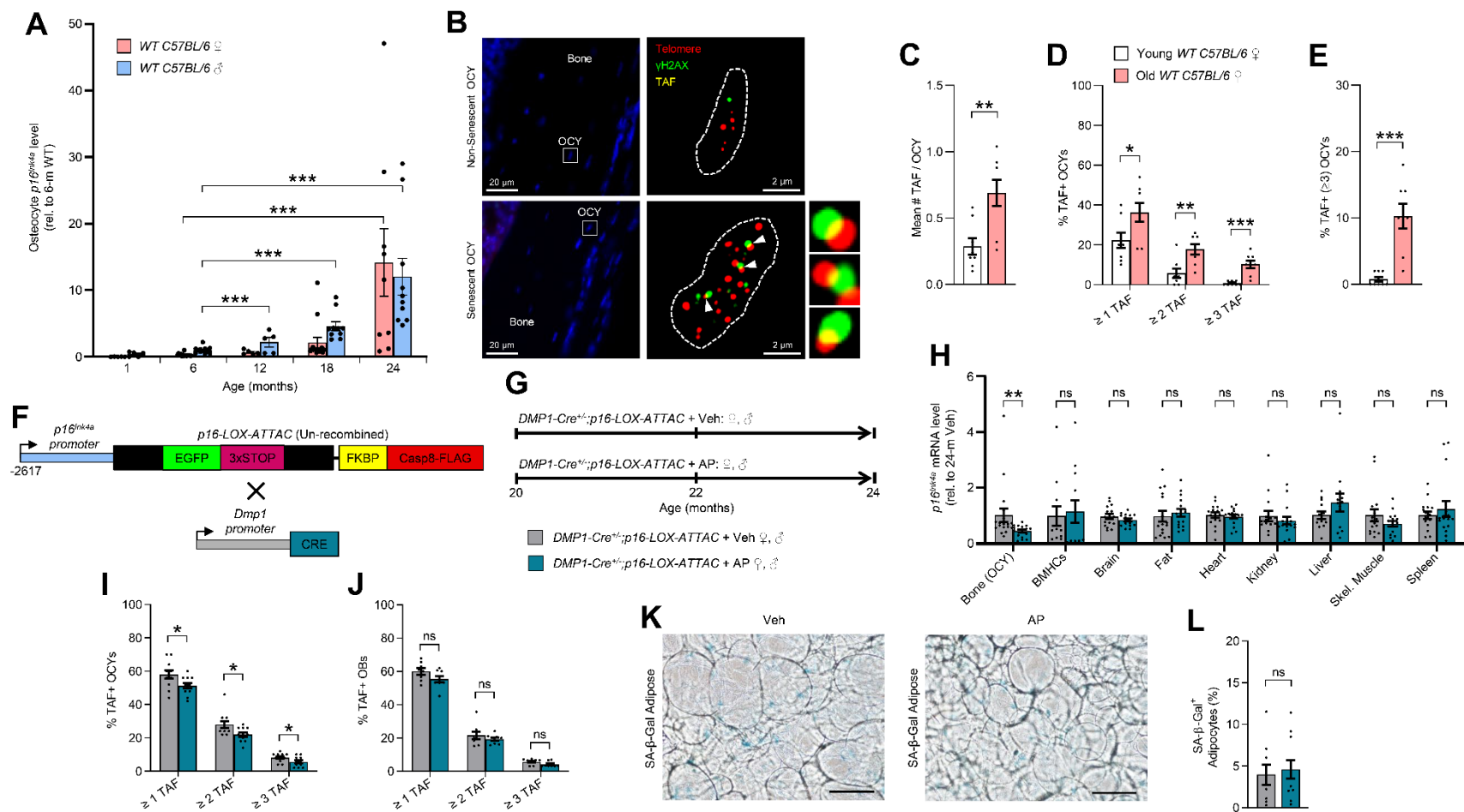


Figure 1. Senescent (Sn) osteocytes (OCYs) accumulate in bone with aging and are cleared by AP20187 treatment in old *DMP1-Cre^{+/-};p16-LOX-ATTAC* mice. (A) Changes in female (♀, pink bars) and male (♂, blue bars) murine *p16^{Ink4a}* mRNA expression throughout life in OCY-enriched bone from C57BL/6 Wild-Type (WT) mice; relative to young adult (6-mo) mice (ANOVA, Tukey *post hoc* test). (B) Representative images of a non-Sn OCY (original magnification, 63x oil) in a young (6-mo) mouse vs a Sn OCY in an old (24-mo) mouse according to the telomere-associated foci (TAF, see white arrows) assay (n=8♀/group). (C-E)

Quantification of **(C)** mean TAF/OCY and **(D)** mean percentage (%) TAF⁺ OCYs/mouse based on: % OCYs ≥ 1 TAF, % OCYs ≥ 2 TAF, % OCYs ≥ 3 TAF, respectively; **(E)** mean % of ≥ 3 TAF⁺ OCYs/mouse (n=8♀/group). **(F)** Schematic of un-recombined *p16-LOX-ATTAC* construct and cross with *DMP1-Cre* mice. **(G)** Study design for local clearance of Sn osteocytes in old (20-mo) *p16-LOX-ATTAC* x *DMP1-Cre* mouse cohorts, males and females combined, randomized to vehicle (Veh, grey) or AP20187 (AP, teal) for four months. **(H)** RT-qPCR analysis of *p16^{Ink4a}* mRNA expression across tissues in mice (males and females combined, n=12-18 per tissue) treated with Veh (grey) vs AP (teal). **(I-J)** Quantification of **(I)** mean % of TAF⁺ OCYs/mouse and **(J)** mean % of TAF⁺ osteoblasts (OBs)/mouse based on: % of cells ≥ 1 TAF, % ≥ 2 TAF, % ≥ 3 TAF, respectively. **(K)** Representative images of perigonadal adipose tissue staining for senescence-associated β -galactosidase (SA- β -Gal)⁺ cells in Veh- and AP-treated mice (Bar = 100 μ m). **(L)** Quantification of SA- β -Gal⁺ adipocytes in mice treated with Veh (grey, n=9; 5♀, 4♂) vs AP (teal, n=11; 6♀, 5♂). Data represent mean SEM $\pm$ (error bars). ns = not significant; **P*<0.05; ***P*<0.01; ****P*<0.001 (independent samples *t*-Test or Wilcoxon's rank-sum test, as appropriate).

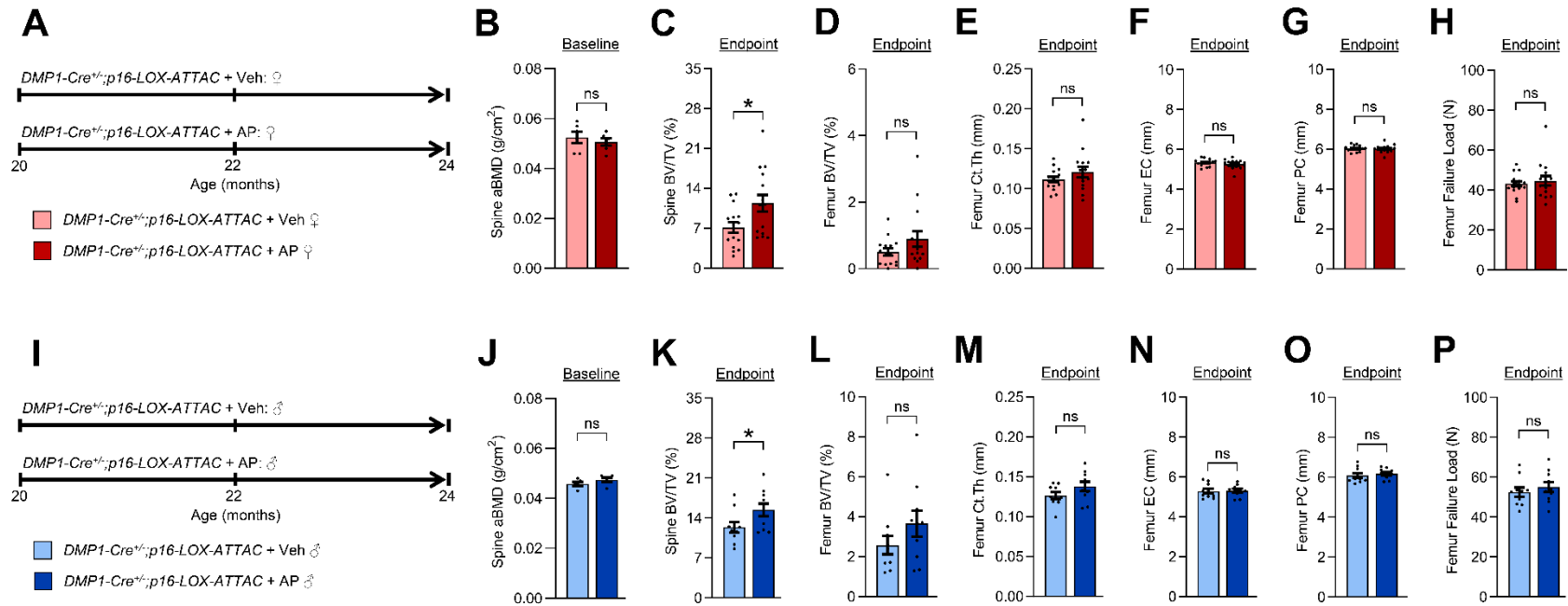


Figure 2. Effects of local senescent (Sn) osteocyte-specific clearance on the skeleton of female and male old mice. (A) Study design for local clearance of Sn osteocytes in female old (20-mo) *DMP1-Cre^{+/+};p16-LOX-ATTAC* mice randomized to Vehicle (Veh, pink) or AP20187 (AP, red) for four months. (B) DXA-derived areal bone mineral density (aBMD, g/cm²) at baseline in females (20-mo; n=6♀/group). (C) Quantification of study endpoint (24-mo) μCT-derived bone volume fraction (BV/TV; %) at the lumbar spine in mice treated with Veh (n=15♀) vs AP (n=15♀). (D-H) Quantification of μCT-derived (D) BV/TV, (E) cortical thickness (Ct.Th), (F) endocortical circumference (EC), (G) periosteal circumference (PC), and (H) μFEA-derived failure load at the femur metaphysis in female mice (n=15/group). (I) Study design for local clearance of Sn osteocytes in male old (20-mo) *DMP1-Cre^{+/+};p16-LOX-ATTAC* mice randomized to Veh (light blue) or AP (dark blue) for four months. (J) DXA-derived aBMD (g/cm²) at baseline in males (20-mo; n=5♂/group). (K) Quantification of study endpoint (24-mo) μCT-derived BV/TV (%) at the lumbar spine in male mice treated with Veh (n=10♂) vs AP (n=10♂). (L-P) Quantification of μCT-derived (L) BV/TV, (M) Ct.Th, (N) EC, (O) PC, and (P) μFEA-derived failure load at the femur metaphysis in male mice (n=10/group). Data represent mean SEM ± (error bars). ns = not significant ($P>0.05$); * $P<0.05$; ** $P<0.01$ (independent samples *t*-Test or Wilcoxon's rank-sum test, as appropriate).

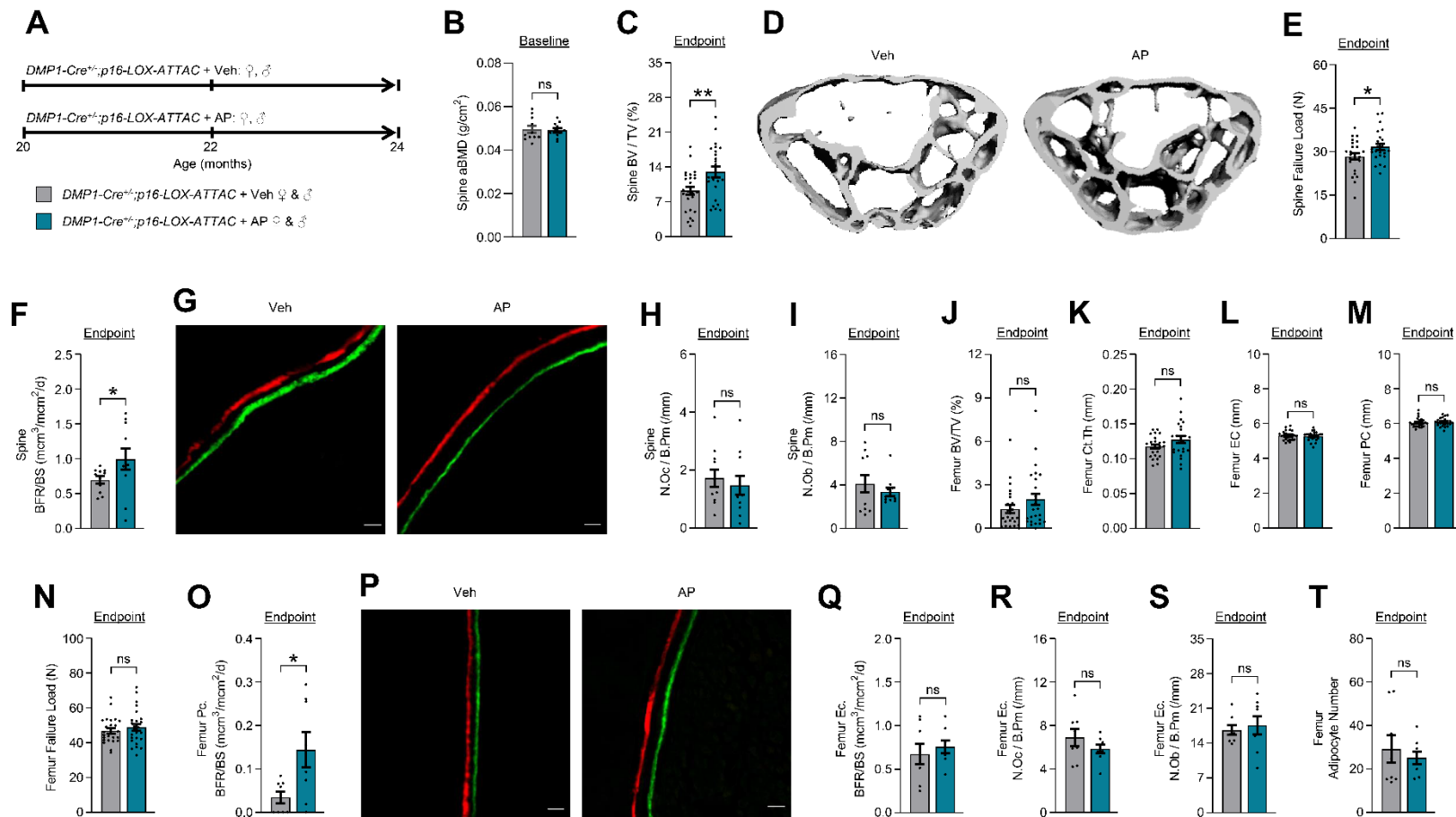


Figure 3. Effects of local senescent (Sn) osteocyte (OCY)-specific clearance on the skeleton of old mice. (A) Study design in old (20-mo) *DMP1-Cre^{+/+};p16-LOX-ATTAC* mice, males and females combined, randomized to Vehicle (Veh, grey) or AP20187 (AP, teal). (B) DXA-derived areal bone mineral density (aBMD, g/cm²) at baseline (20-mo; n=11/group; 6♀, 5♂ per group). (C) Endpoint lumbar spine μ CT-derived bone volume fraction (BV/TV; %) in Veh- (n=25; 15♀, 10♂) vs AP-treated (n=25; 15♀, 10♂) mice. (D) Representative spine μ CT images of Veh- vs AP-treated mice. (E) Lumbar spine μ -finite-element analysis (μ FEA)-derived failure load in Veh- (n=25; 15♀, 10♂) vs AP-treated (n=25; 15♀, 10♂) mice. (F) Lumbar spine bone formation rate [BFR] per bone surface (BFR/BS; mcm³/mcm²/d; n=11/group; 6♀, 5♂ per group). (G) Lumbar spine representative images of alizarin red and calcein

fluorochrome dynamic BFR labels on trabecular surfaces in Veh- and AP-treated mice (scale bar = 10 μ m). **(H-I)** Lumbar spine histomorphometric quantification (n=8/group; 4♀, 4♂ per group) of **(H)** osteoclast numbers per bone perimeter (N.Oc/B.Pm; /mm) and **(I)** osteoblast numbers per bone perimeter (N.Ob/B.Pm; /mm). **(J-N)** Femur metaphysis μ CT-derived **(J)** BV/TV, **(K)** cortical thickness (Ct.Th), **(L)** endocortical circumference (EC), **(M)** periosteal circumference (PC), and **(N)** μ FEA-derived failure load (n=25/group; 15♀, 10♂ per group). **(O)** Femur histomorphometric quantification (n=8/group; 4♀, 4♂ per group) of periosteal cortical BFR/BS ($\text{mcm}^3/\text{mcm}^2/\text{d}$). **(P)** Femur representative images of alizarin red and calcein fluorochrome dynamic BFR labels on periosteal cortical (Pc.) surfaces in Veh- and AP-treated mice (scale bar = 10 μ m). **(Q-T)** Femur histomorphometric quantification (n=8/group; 4♀, 4♂ per group) of **(Q)** endocortical (Ec.) BFR/BS ($\text{mcm}^3/\text{mcm}^2/\text{d}$), **(R)** Ec. N.Oc/B.Pm (/mm), **(S)** Ec. N.Ob/B.Pm (/mm), and **(T)** bone marrow adipocyte numbers. Data represent mean SEM $\pm$ (error bars). ns = not significant ($P>0.05$); * $P<0.05$; ** $P<0.01$ (independent samples *t*-Test or Wilcoxon's rank-sum test, as appropriate).

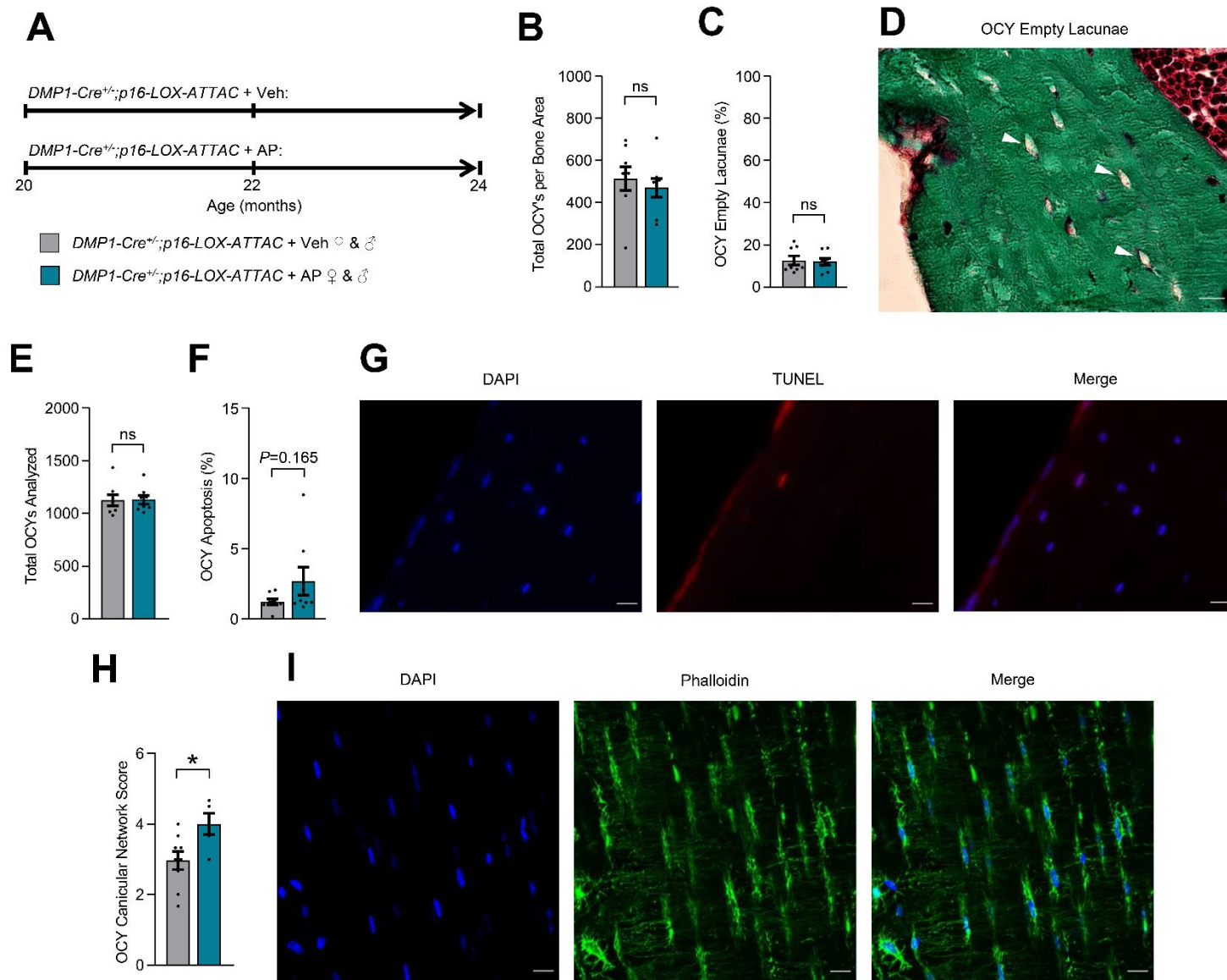


Figure 4. Effects of local senescent (Sn) osteocyte (OCY) clearance on empty lacunae, OCY apoptosis, and OCY canicular network quality. (A) Study design for the local clearance of Sn osteocytes in old (20-mo) *DMP1-Cre^{+/+};p16-LOX-ATTAC* mouse cohorts, males and females combined, randomized to Vehicle (Veh, grey) or AP20187 (AP, teal) for four months. (B-D) Study endpoint (24-mo) quantification of (B) total OCY's per bone area and (C) percentage (%) of OCY empty lacunae in mice treated with Veh (n=8; 4♀, 4♂) vs AP (n=8; 4♀, 4♂); (D) representative images of OCY empty lacunae (scale bar = 25 µm). (E-G) Quantification of OCY apoptosis, including (E) total numbers of OCYs analyzed and (F) % of OCY apoptosis in mice treated with Veh (n=8; 4♀, 4♂) vs AP (n=8; 4♀, 4♂); (G) representative images of DAPI-stained, TUNEL⁺, and merged apoptotic OCYs (scale bar = 25 µm). (H) Quantification of OCY lacunocanicular network (LCN) score in mice treated with Veh (n=9; 5♀, 4♂) vs AP (n=5, 3♀, 2♂); (I) representative images of DAPI-stained, phalloidan-stained, and merged OCY LCN (scale bar = 25 µm). Data represent mean SEM ± (error bars). ns = not significant ($P>0.05$); * $P<0.05$ (independent samples *t*-Test or Wilcoxon's rank-sum test, as appropriate).

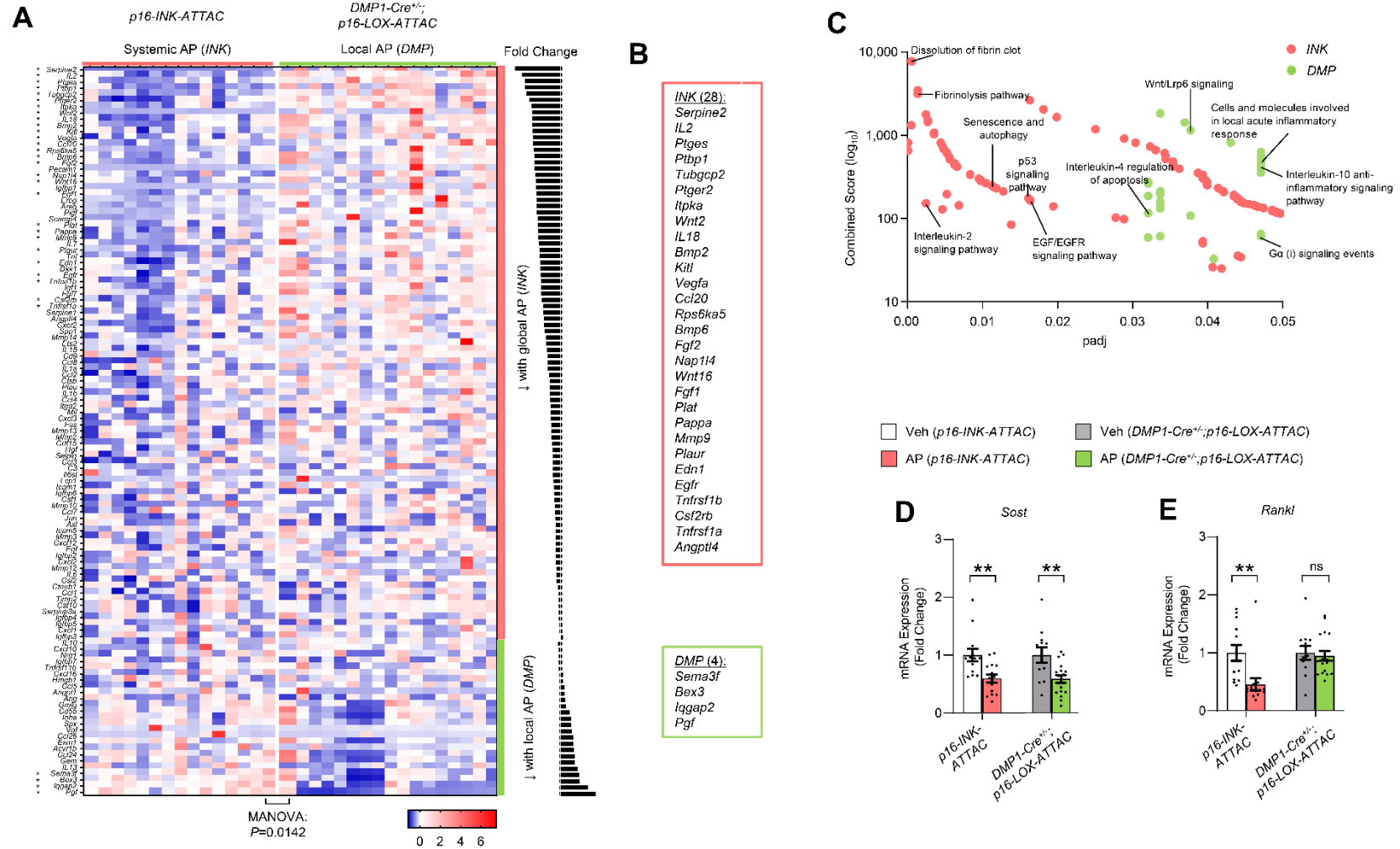


Figure 5. Changes in SenMayo genes and regulators of bone remodeling in response to local osteocyte-specific (*DMP*) versus systemic (*INK*) senescent cell (SnC) clearance. (A-B) Following 2-wks, comparison of the two AP20187 (AP)-treated groups (*INK* versus *DMP*) revealed broader downregulation of SASP and Sn-related genes in the systemic SnC clearance model (*INK*) vs the local osteocyte-specific model (*DMP*). **(C)** BioPlanet2019 pathway analysis emphasized the senescence and autophagy as well as the p53/IL2/Egf pathways as regulated in the *INK*-model, whereas the Wnt/Lrp6/IL10/IL4 pathways were regulated in the

DMP-model. **(D-E)** Changes in mRNA expression of **(D)** *Sost* (encoding sclerostin) and **(E)** *Rankl* (also known as *Tnfsf11*) in response to systemic (*INK*) vs local osteocyte-specific (*DMP*) senolysis (*INK* Veh: n=15, *INK* AP: n=15; *DMP* Veh: n=15, *DMP* AP: n=15). ns = not significant ($P>0.05$); ** $P<0.01$ (independent samples *t*-Test or Wilcoxon's rank-sum test, as appropriate).

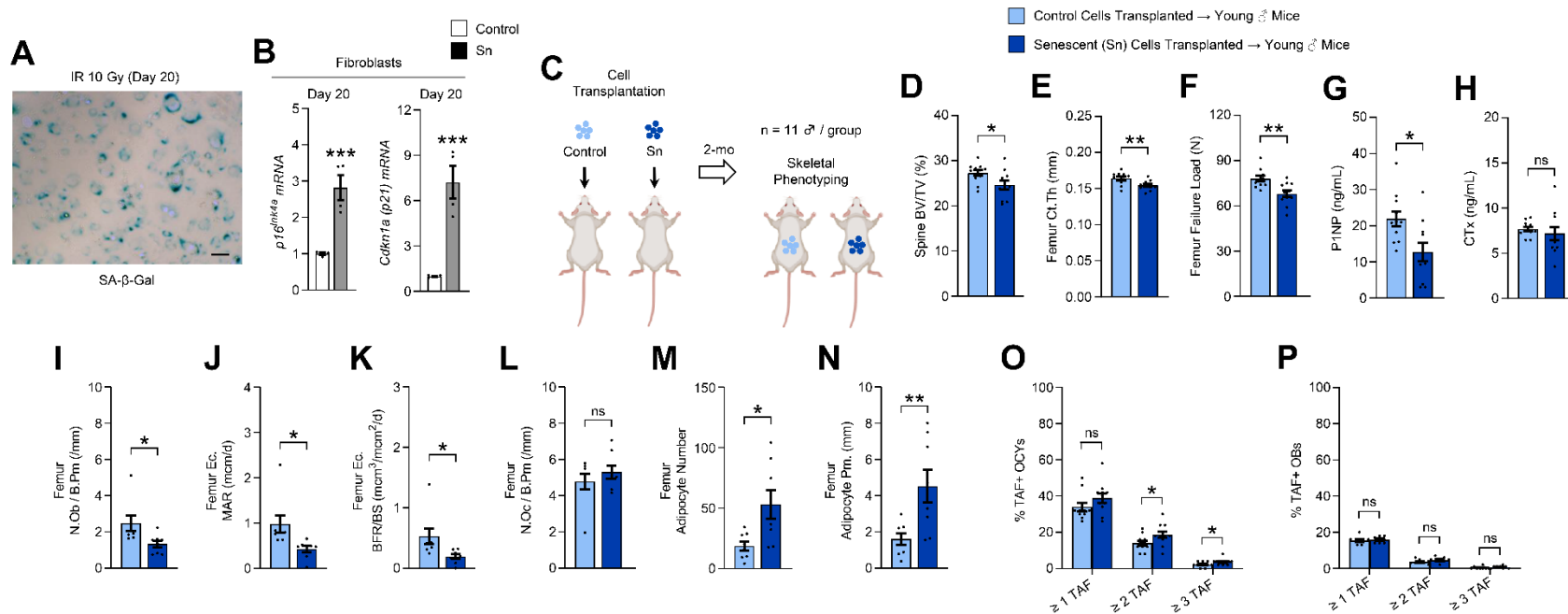


Figure 6. Transplanting cells made senescent (Sn) by irradiation (IR) causes skeletal aging in young adult mice. (A) Confirmation of cellular senescence 20 days following 10 Gy IR using senescence-associated β -galactosidase (SA- β -Gal) staining (scale bar = 25 μ m). **(B)** RT-qPCR analysis of *p16^{Ink4a}* and *p21^{Cip1}* mRNA expression in Sn versus Control fibroblasts 20 days following 10 Gy IR. **(C)** Study design for Control and Sn cell transplantation (via intraperitoneal [i.p.] injection) and skeletal phenotyping in young adult C57BL/6 Wild-Type (WT) mice; n=11♂/group. **(D-F)** Quantification of μ CT-derived **(D)** lumbar spine bone volume fraction (BV/TV; %), **(E)** femur metaphysis cortical thickness (Ct.Th), and **(F)** femur metaphysis μ -finite-element analysis (μ FEA)-derived failure load (i.e. bone strength) in mice transplanted with Control (n=11♂) vs Sn (n=11♂) cells. **(G-H)** Quantification of circulating serum bone turnover markers, including **(G)** the bone formation marker, amino-terminal propeptide of type I collagen (P1NP; ng/mL) and **(H)** the bone resorption marker cross-linked C-telopeptide of type I collagen (CTX; ng/mL); n=11♂/group. **(I-L)** Histomorphometric quantification at the femoral endocortical surface of **(i)** osteoblast numbers per bone perimeter (N.Ob/B.Pm; /mm), **(J)** mineral apposition rate (MAR; mcm/d), **(K)** bone formation rate per bone surface (BFR/BS; mcm³/mcm²/d), and **(L)** osteoclast numbers per bone perimeter (N.Oc/B.Pm; /mm); n=8♂/group. **(M-N)** Histomorphometric quantification of femur bone marrow **(M)** adipocyte numbers and **(N)** adipocyte perimeter (Pm; mm); n=8♂/group. **(O-P)** Quantification of **(P)** the mean percentage

(%) of TAF⁺ OCYs per mouse and (**P**) the mean % of TAF⁺ osteoblasts (OBs) per mouse based on the following criteria: % of cells with ≥ 1 TAF, % with ≥ 2 TAF, % with ≥ 3 TAF, respectively. Data represent mean SEM $\pm$ (error bars). ns = not significant ($P > 0.05$); * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$ (independent samples *t*-Test or Wilcoxon's rank-sum test, as appropriate).

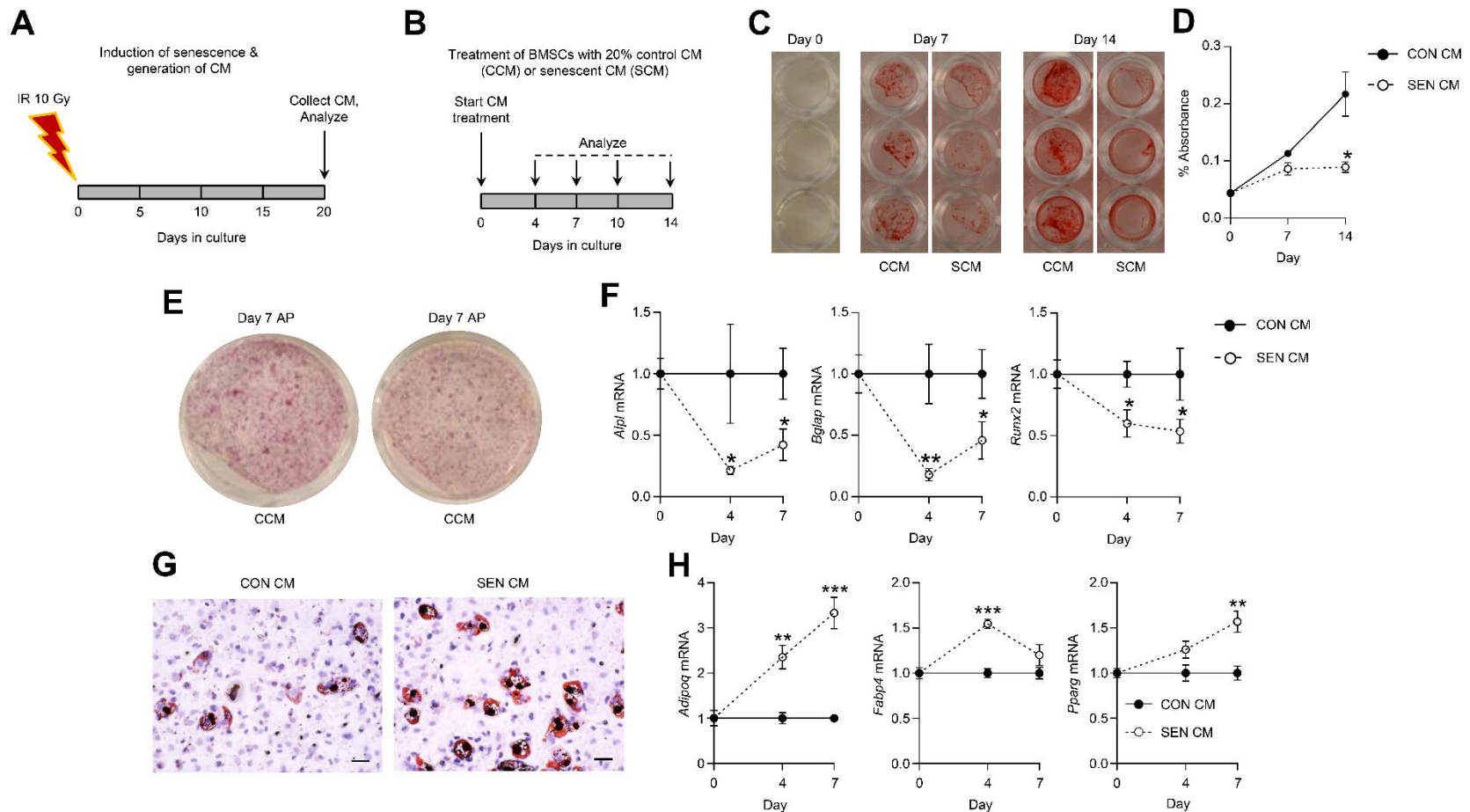


Figure 7. The senescence-associated secretory phenotype (SASP) impairs bone formation and enhances bone marrow adipogenesis. (A) Schematic of *in vitro* cellular senescence induction using 10 Gy irradiation (IR) and generation of conditioned medium (CM) following 20 days in culture. (B) Study design depicting treatment of bone marrow stromal cells (BMSCs) with 20% control CM (CON CM) or senescent CM (SEN CM) for 14 days. (C) Mineralization of BMSCs exposed to CON CM or SEN CM at days 0, 7 (n=6/group), and 14 (n=6/group). (D) Quantification of eluted alizarin red dye from BMSCs exposed to CON CM or SEN

CM at days 0, 7 (n=6/group), and 14 (n=6/group). **(E)** Representative images of alkaline phosphatase (AP) staining in BMSCs exposed to CON CM or SEN CM at day 7. **(F)** Changes in mRNA expression of osteoblastic genes in BMSCs exposed to either CON CM (n=6) or SEN CM (n=6). **(G)** Representative images of Oil Red O staining of BMSCs exposed to either CON CM or SEN CM (scale bar = 25 μ m). **(H)** Changes in mRNA expression of adipogenic genes in BMSCs exposed to either CON CM or SEN CM; n=6/group. Data represent mean SEM $\pm$ (error bars). * P <0.05; ** P <0.01; *** P <0.001 (independent samples t -Test or Wilcoxon's rank-sum test, as appropriate).